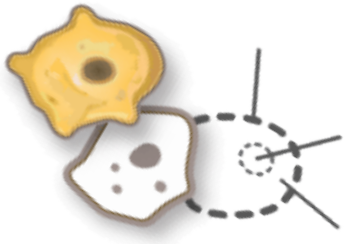
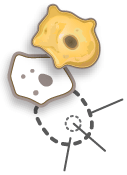
Web Services and Development of Semantic Applications

Trish Whetzel

Outreach Coordinator

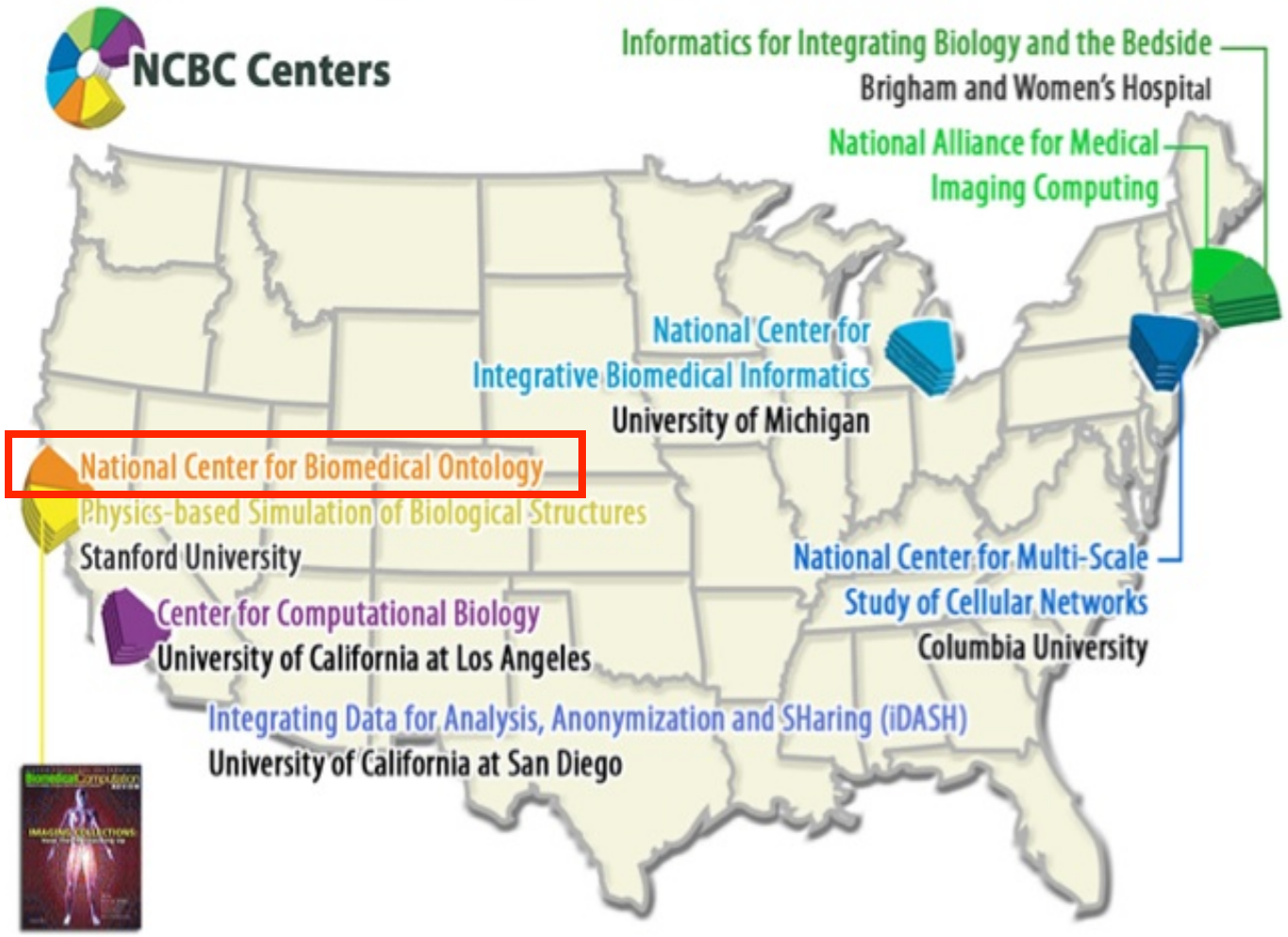


THE NATIONAL CENTER FOR
BIOMEDICAL ONTOLOGY



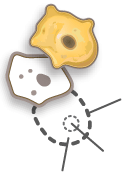
National Center for Biomedical Ontology

- Mission
  - To create software for the application of ontologies in biomedical science and clinical care
- NCBO Partners
  - Stanford University - Dr. Mark A. Musen
  - Mayo Clinic - Dr. Christopher G. Chute
  - University of Buffalo - Dr. Barry Smith
  - University of Victoria - Dr. Margaret-Anne Storey



National Centers for Biomedical Computing

(http://www.ncbcs.org)



NCBO Key Activities

- We ***create and maintain a library*** of biomedical ontologies
- We ***build tools and Web services*** to enable the use of ontologies
- We ***collaborate with scientific communities*** that develop and use ontologies

THE NATIONAL CENTER FOR
BIOMEDICAL ONTOLOGY

Search | Search Site | Search Google | LOGIN

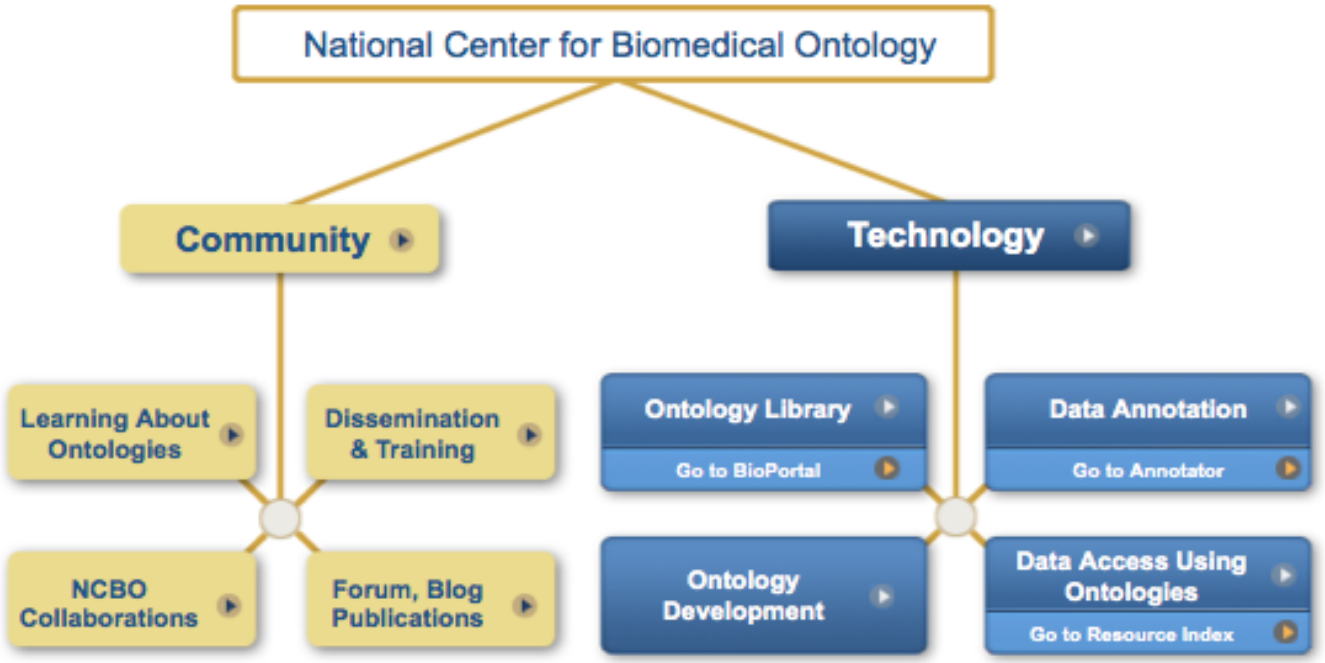
Home | News & Events | Dissemination & Training | Collaboration | Community | Technology | About Us

Like 111

Of Current Interest

- **News**: Study Extracts Vioxx Heart-Attack Risk from EMRs, Shift from Ontology Development to Use
- **Webinar**: Wed, Dec. 7, 10:00am PT, Stephen Wu, Mayo Clinic, UMLS Term Occurrences in Clinical Notes: A Large Scale Corpus Analysis
- **Recent Publication**: Whetzel, PL, et al. (2011): BioPortal: enhanced functionality via new Web services...
- **Recent Release**: BioPortal 3.4 (Oct 31, 2011)
- **NCBO Webinar Announcements - Subscribe**
- **NCBO Software Support - Mailing List Archive**
- **More News & Events**

Follow us on Twitter



NCBO User Profile

Susanna-Assunta Sansone, Eamonn Maguire, Philippe Rocca-Serra
University of Oxford



"Our job is to contribute to the realization of the data sharing vision..."

More >

Other profiles >

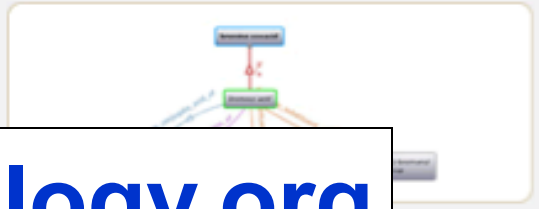
Video

Learn about Biomedical Ontologies. Watch a series of introductory videos.

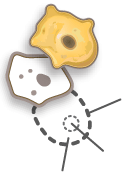


Browse ontologies in BioPortal!

BioPortal allows users to browse, search and visualize ontologies.

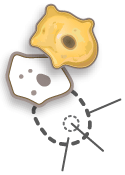


www.bioontology.org



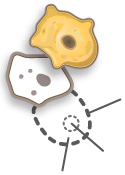
Links of Interest

- http://www.bioontology.org/wiki/index.php/SWAT4LS_Tutorial



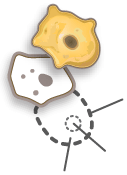
Outline

- Learning Objectives
- REST Web services
- BioPortal
- NCBO Web Services
- BioPortal SPARQL Endpoint



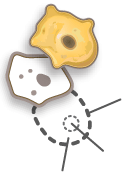
Learning Objectives

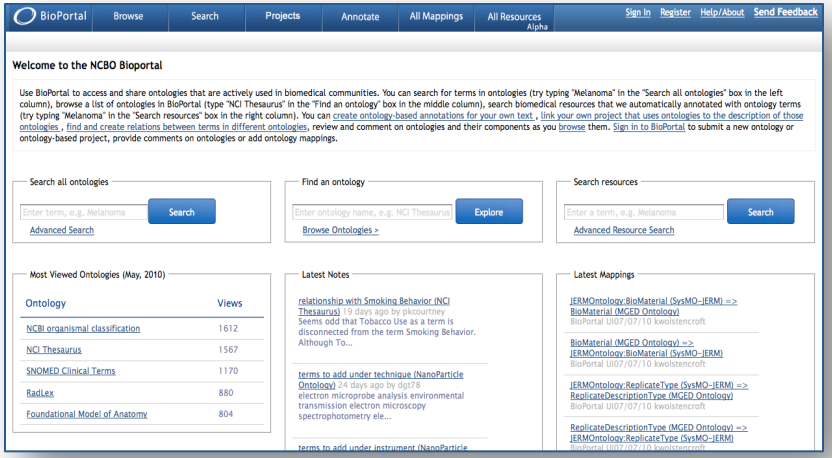
- Learn what Web services are available from NCBO
- Learn how to programmatically use these Web services
- Learn how to combine these Web services to perform tasks

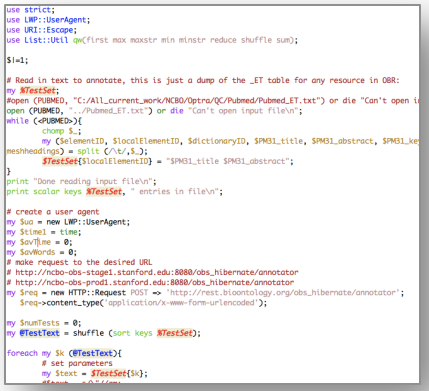


REST Web Services

- Accessed via HTTP
  - http://rest.bioontology.org/{parameters}
- Each unique URL is a representation of some object
- Operations include GET, POST, PUT, DELETE
- Lightweight, easy to build







```
use strict;
use LWP::UserAgent;
use URI::Escape;
use List::Util qw(first max maxstr min minstr reduce shuffle sum);

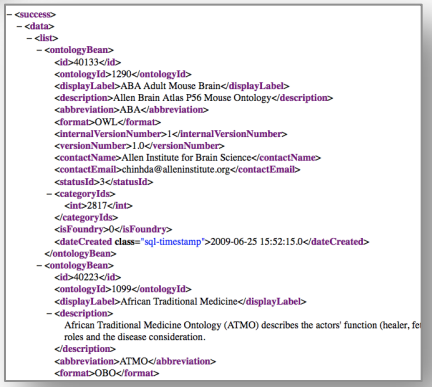
$|=1;

# Read in text to annotate, this is just a dump of the _ET table for any resource in OBR:
my %TestSet;
#open (PUBMED, "C:/All_current_work/NCBO/Optra/QC/Pubmed/Pubmed_ET.txt") or die "Can't open i
open (PUBMED, "../Pubmed_ET.txt") or die "Can't open input file\n";
while (<PUBMED>){
        chomp $_;
        my ($elementID, $localElementID, $dictionaryID, $PM31_title, $PM31_abstract, $PM31_ke
meshheadings) = split (/\t/,$_);
        $TestSet{$localElementID} = "$PM31_title $PM31_abstract";
}
print "Done reading input file\n";
print scalar keys %TestSet, " entries in file\n";

# create a user agent
my $ua = new LWP::UserAgent;
my $time1 = time;
my $avTime = 0;
my $avWords = 0;
# make request to the desired URL
# http://ncbo-obs-stage1.stanford.edu:8080/obs_hibernate/annotator
# http://ncbo-obs-prod1.stanford.edu:8080/obs_hibernate/annotator
my $req = new HTTP::Request POST => 'http://rest.bioontology.org/obs_hibernate/annotator';
   $req->content_type('application/x-www-form-urlencoded');

my $numTests = 0;
my @TestText = shuffle (sort keys %TestSet);

foreach my $k (@TestText){
        # set parameters
        my $text = $TestSet{$k};
```



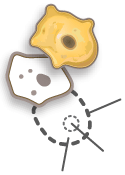
```
- <success>
  - <data>
    - <list>
      - <ontologyBean>
          <id>40133</id>
          <ontologyId>1290</ontologyId>
          <displayLabel>ABA Adult Mouse Brain</displayLabel>
          <description>Allen Brain Atlas P56 Mouse Ontology</description>
          <abbreviation>ABA</abbreviation>
          <format>OWL</format>
          <internalVersionNumber>1</internalVersionNumber>
          <versionNumber>1.0</versionNumber>
          <contactName>Allen Institute for Brain Science</contactName>
          <contactEmail>chinhda@alleninstitute.org</contactEmail>
          <statusId>3</statusId>
        - <categoryIds>
            <int>2817</int>
          </categoryIds>
          <isFoundry>0</isFoundry>
          <dateCreated class="sql-timestamp">2009-06-25 15:52:15.0</dateCreated>
        </ontologyBean>
      - <ontologyBean>
          <id>40223</id>
          <ontologyId>1099</ontologyId>
          <displayLabel>African Traditional Medicine</displayLabel>
        - <description>
            African Traditional Medicine Ontology (ATMO) describes the actors' function (healer, fe
            roles and the disease consideration.
          </description>
          <abbreviation>ATMO</abbreviation>
          <format>OBO</format>
```

User Interface

Code

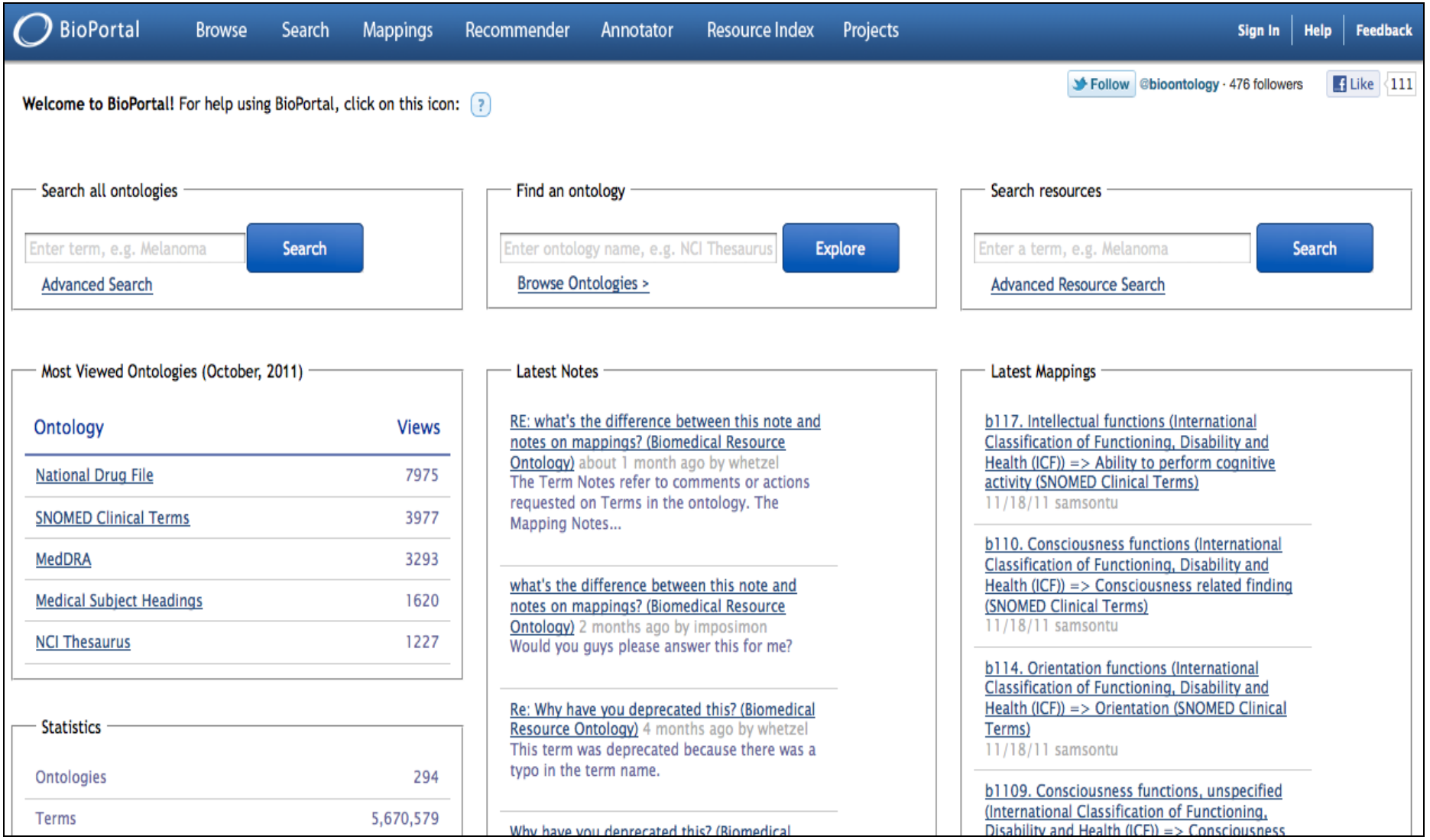
Browser

Web Services

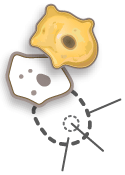


Outline

- REST Web services
- BioPortal
- NCBO Web Services
- BioPortal SPARQL Endpoint

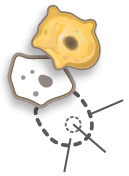


http://bioportal.bioontology.org



Outline

- REST Web services
- BioPortal
- NCBO Web Services
- BioPortal SPARQL Endpoint

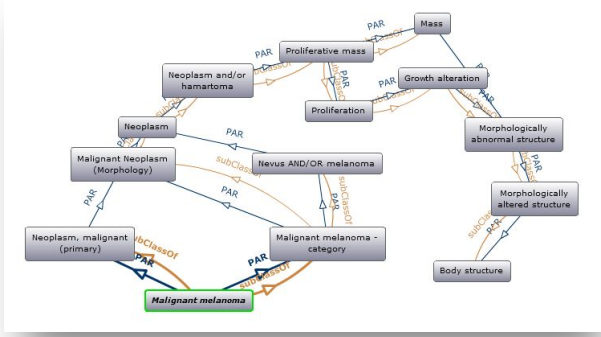


http://rest.bioontology.org

Ontology Services

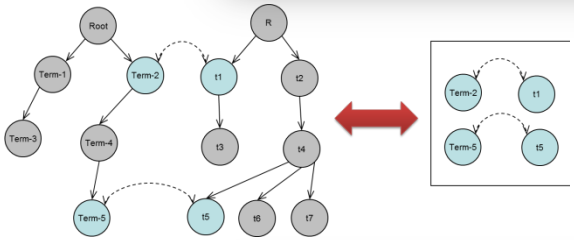
Views

- Search
- Traverse
- Comment
- Download



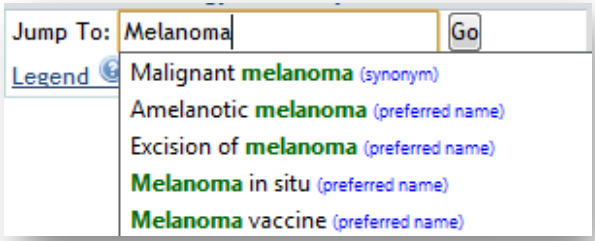
Mapping Services

- Create
- Upload
- Download



Widgets

- Tree-view
- Auto-complete
- Graph-view



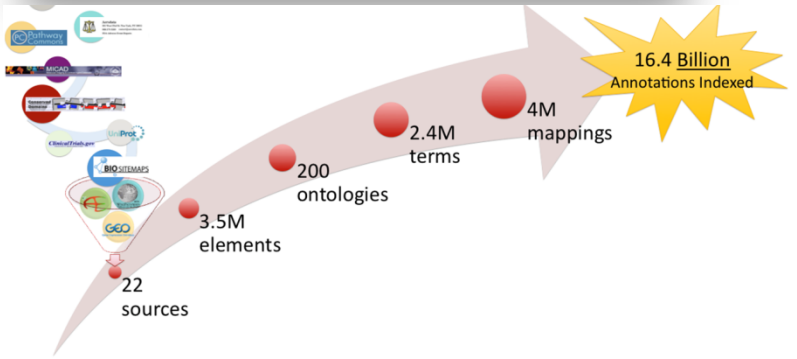
Annotation

Term recognition

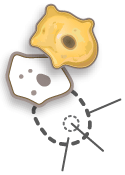
Expression, Expression of bladder, bladder, smooth, bladder muscle, muscle, smooth muscle, cells, mechanical, mechanical stimulation, stimulation, Chronic, results, bladder overdistension, associated, associated with, with, loss, genes, altered

Data Access

Fetch "data" annotated with a given term

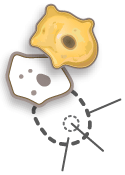


http://bioportal.bioontology.org



ONTOLOGY WEB SERVICES

Accessing, browsing, searching and traversing ontologies in *Your* application



Ontology Web Services

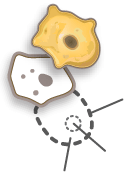
- List All Ontologies
  - List all ontologies in BioPortal
  - Documentation: http://bit.ly/sxL2Qi
  - GoogleDoc Excel example: http://bit.ly/ncbo-list-all-ontologies

```
- <success>
    <accessedResource>/bioportal/ontologies</accessedResource>
    <accessDate>2011-12-04 22:01:39.603 PST</accessDate>
  - <data>
    - <list>
      + <ontologyBean></ontologyBean>
      + <ontologyBean></ontologyBean>
      + <ontologyBean></ontologyBean>
      - <ontologyBean>
          <id>44776</id>
          <ontologyId>1351</ontologyId>
        - <virtualViewIds>
            <int>1420</int>
            <int>1456</int>
          </virtualViewIds>
          <internalVersionNumber>3</internalVersionNumber>
        - <userIds>
            <int>38134</int>
          </userIds>
          <versionNumber>2011_2010_08_30</versionNumber>
          <isRemote>0</isRemote>
          <statusId>3</statusId>
          <dateCreated>2010-11-09 11:16:42.0 PST</dateCreated>
          <dateReleased>2010-08-30 00:00:00.0 PDT</dateReleased>
          <isManual>1</isManual>
          <displayLabel>Medical Subject Headings</displayLabel>
        - <description>
            Medical Subject Headings (MeSH);National Library of Medicine;Februrary, 2009;Bethesda, MD;ENG
          </description>
          <abbreviation>MSH</abbreviation>
          <format>RRF</format>
```


One ontology id has many ontology version ids

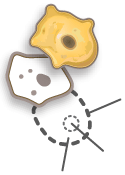
Ontology version id

Ontology id



Ontology Web Services cont.

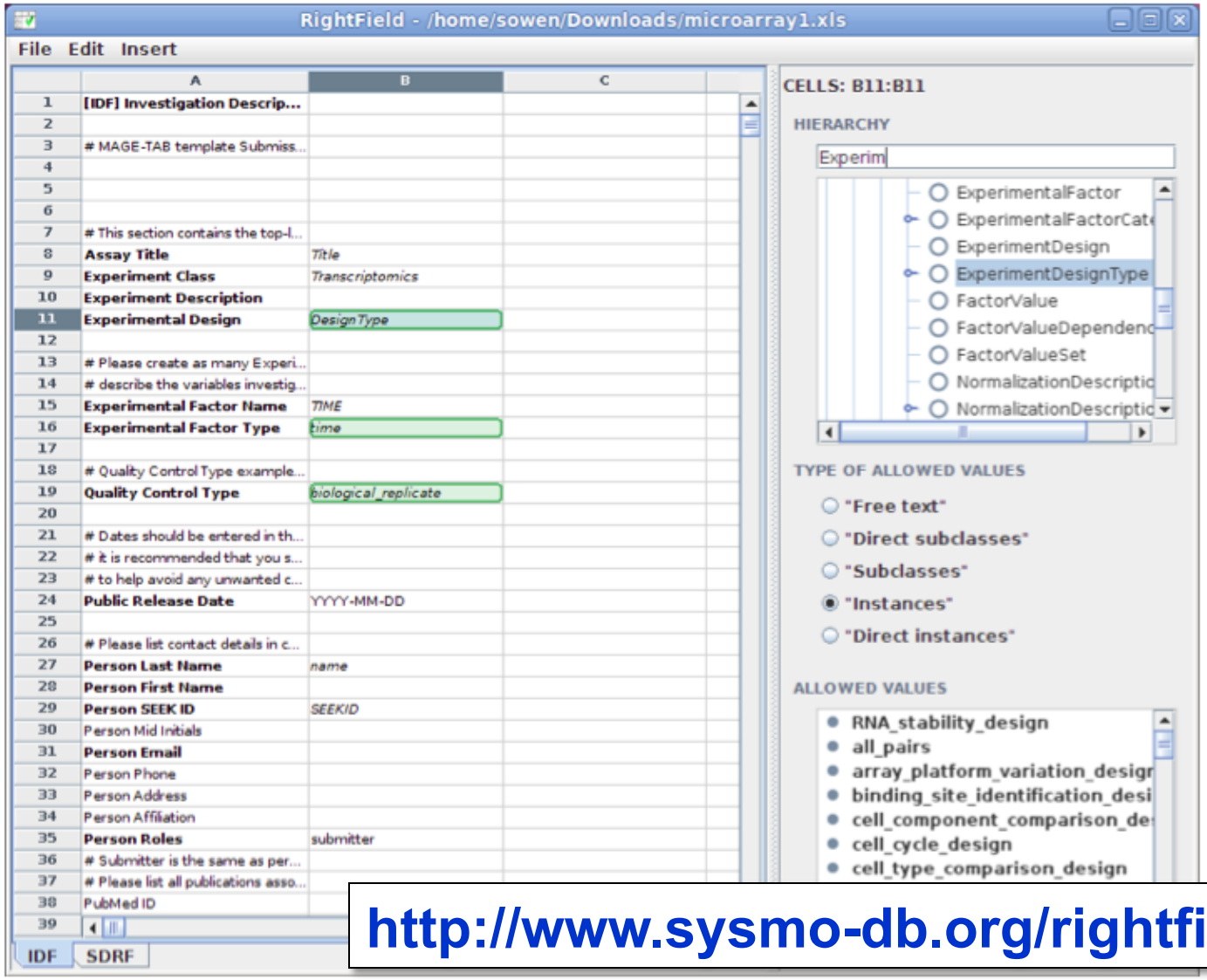
- Search
  - Search by term or term identifier across all ontologies in BioPortal
- Documentation: http://bit.ly/tMDkr6
- Example clients
  - Java: http://bit.ly/ttTb1G
  - Perl: http://bit.ly/sPXC2X



Ontology Web Services cont.

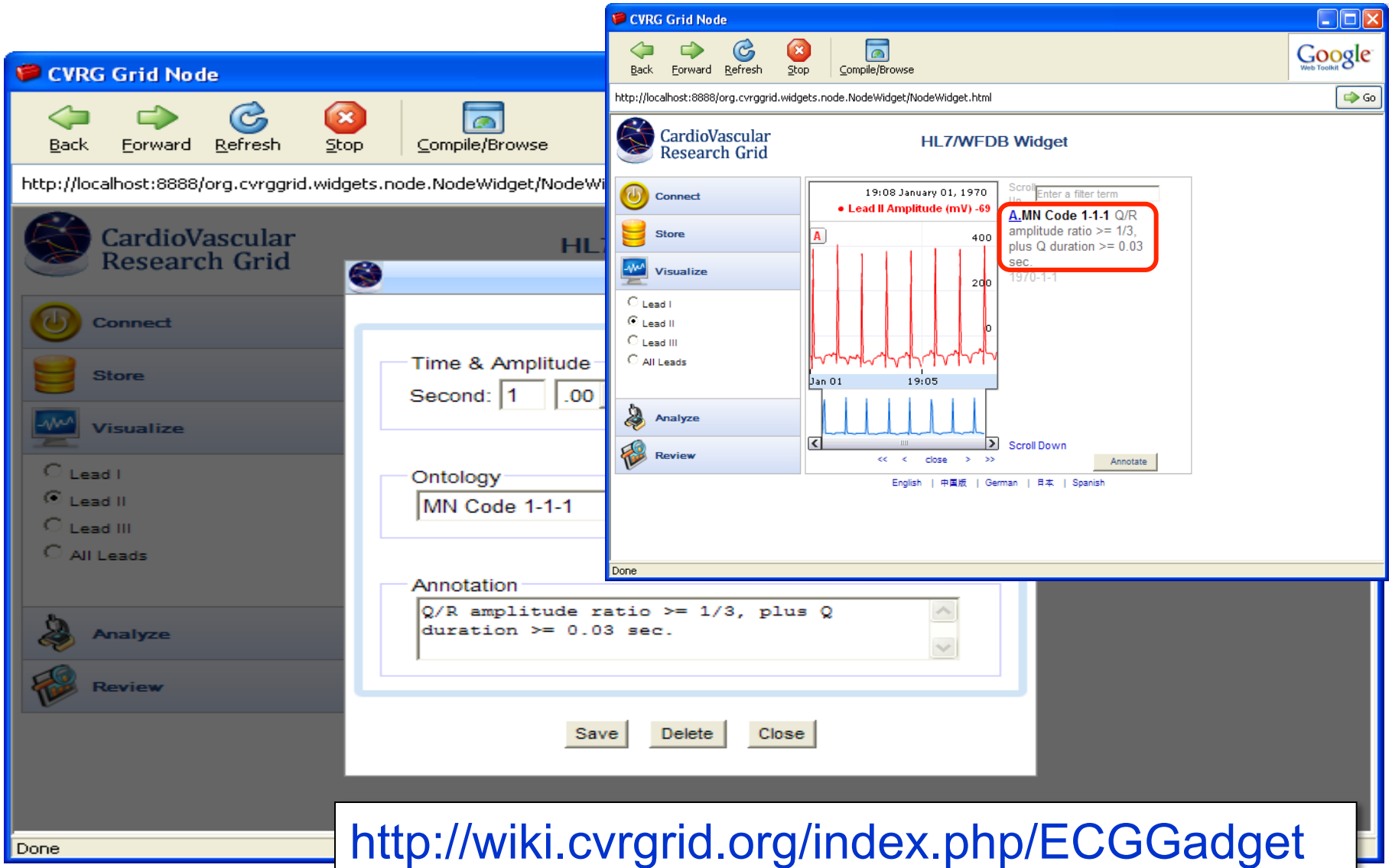
- Term
  - get details about a term including it's definition, synonyms, URI, super- and sub-classes, and other properties http://bit.ly/teedbd
- Hierarchy
  - get parent, children, and sibling terms http://bit.ly/tPzm75

RightField



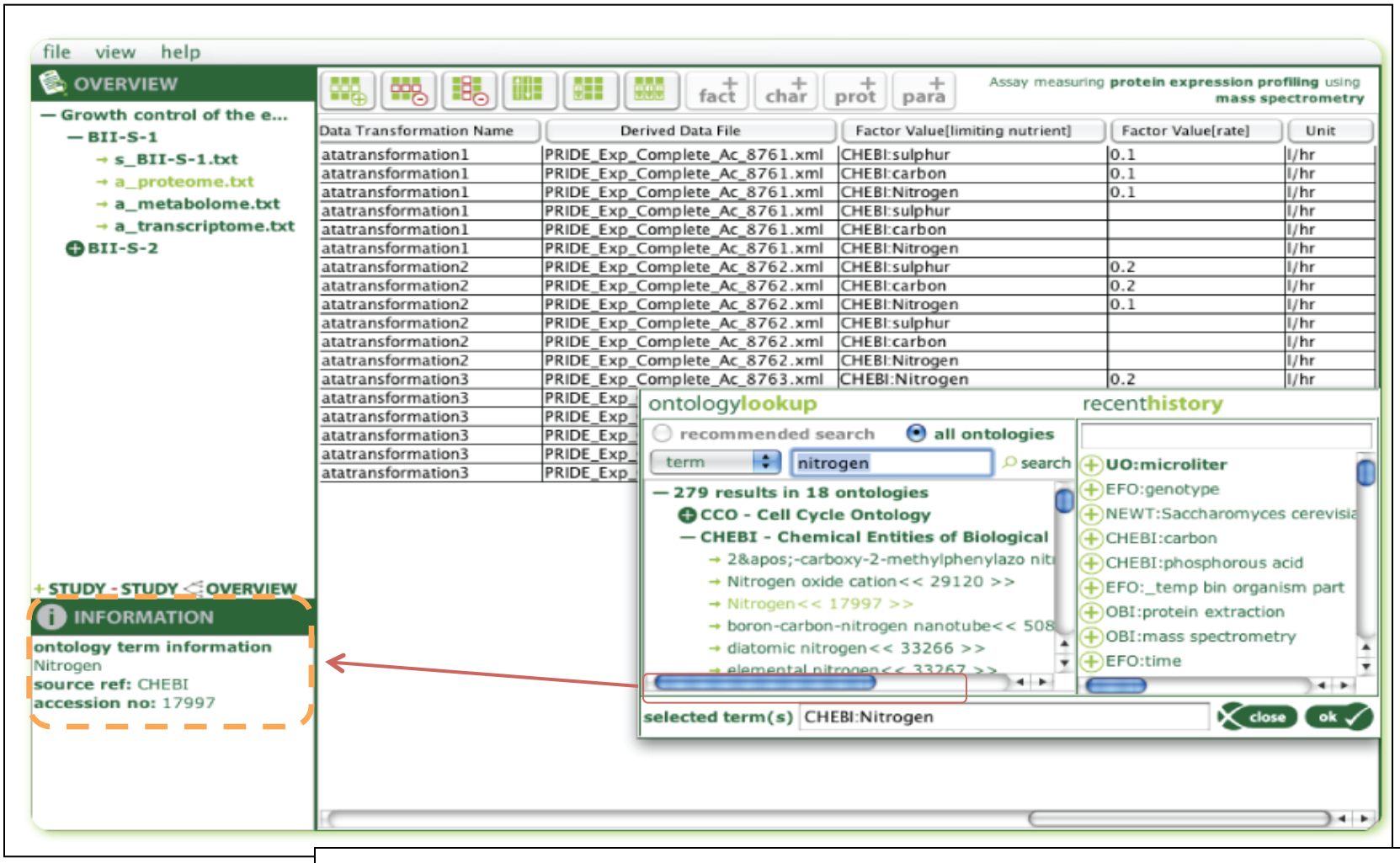
http://www.sysmo-db.org/rightfield

ECG Gadget



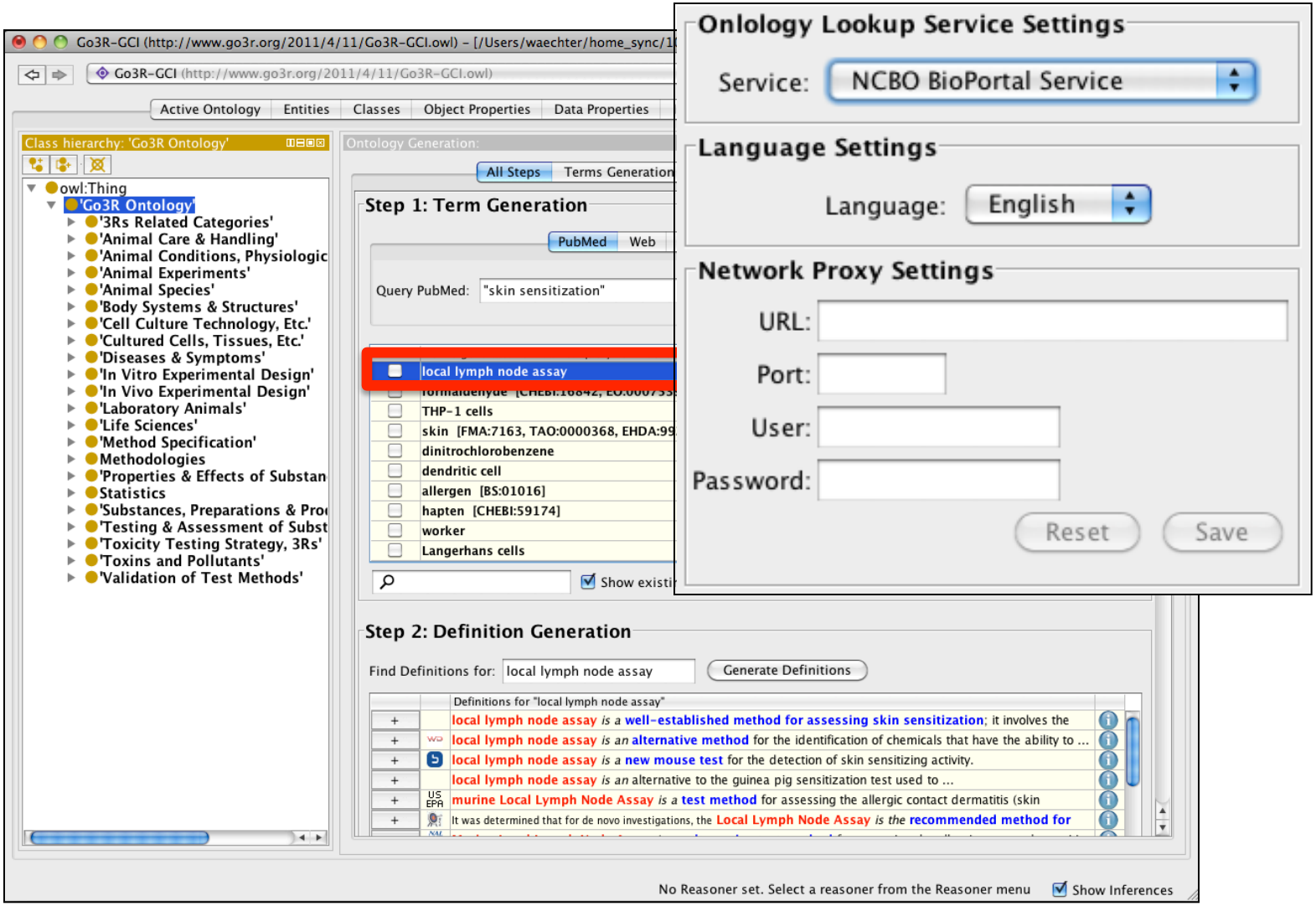
http://wiki.cvrgrid.org/index.php/ECGGadget

ISAcreator

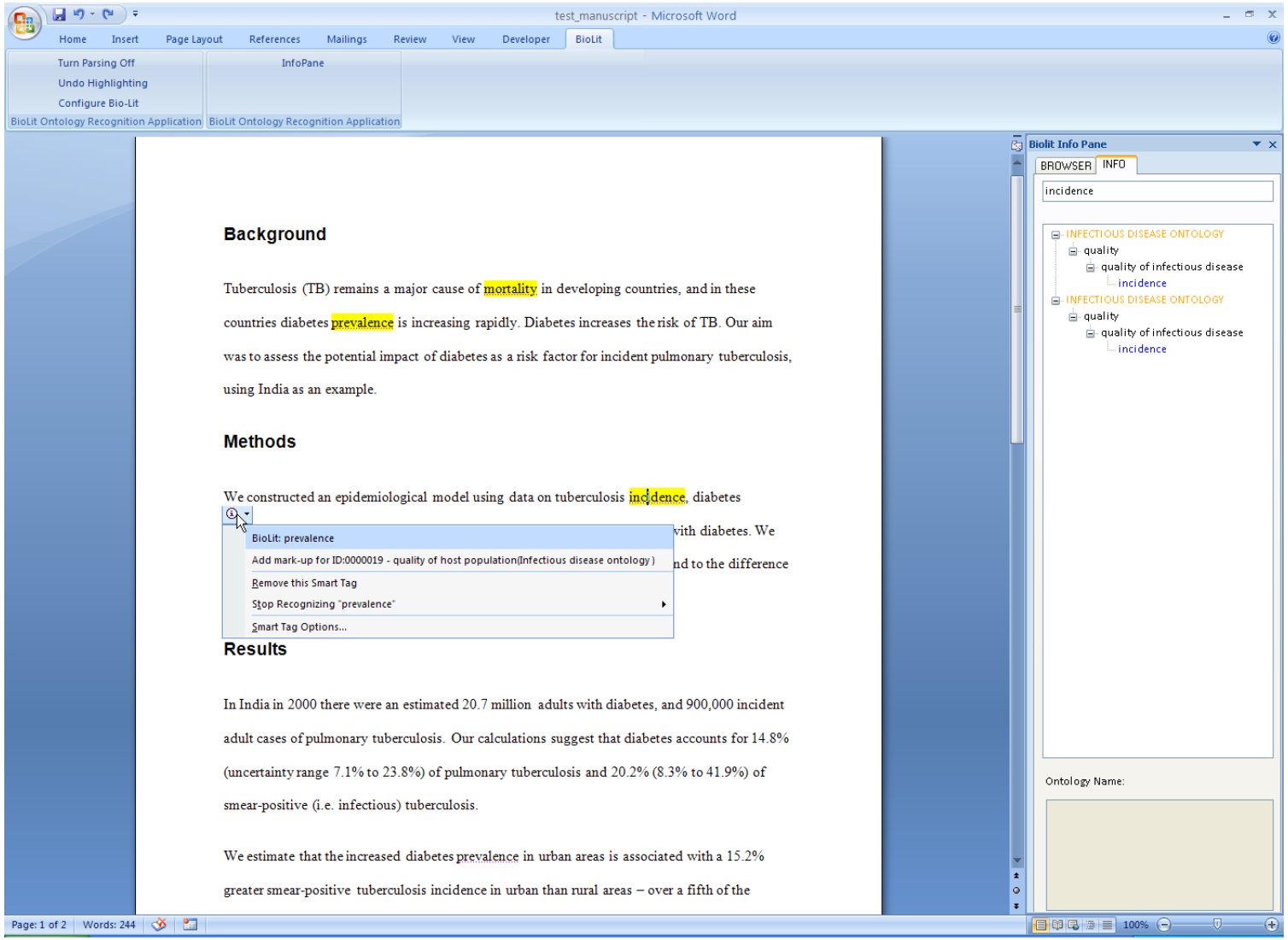


http://isatab.sourceforge.net/isacreator.html

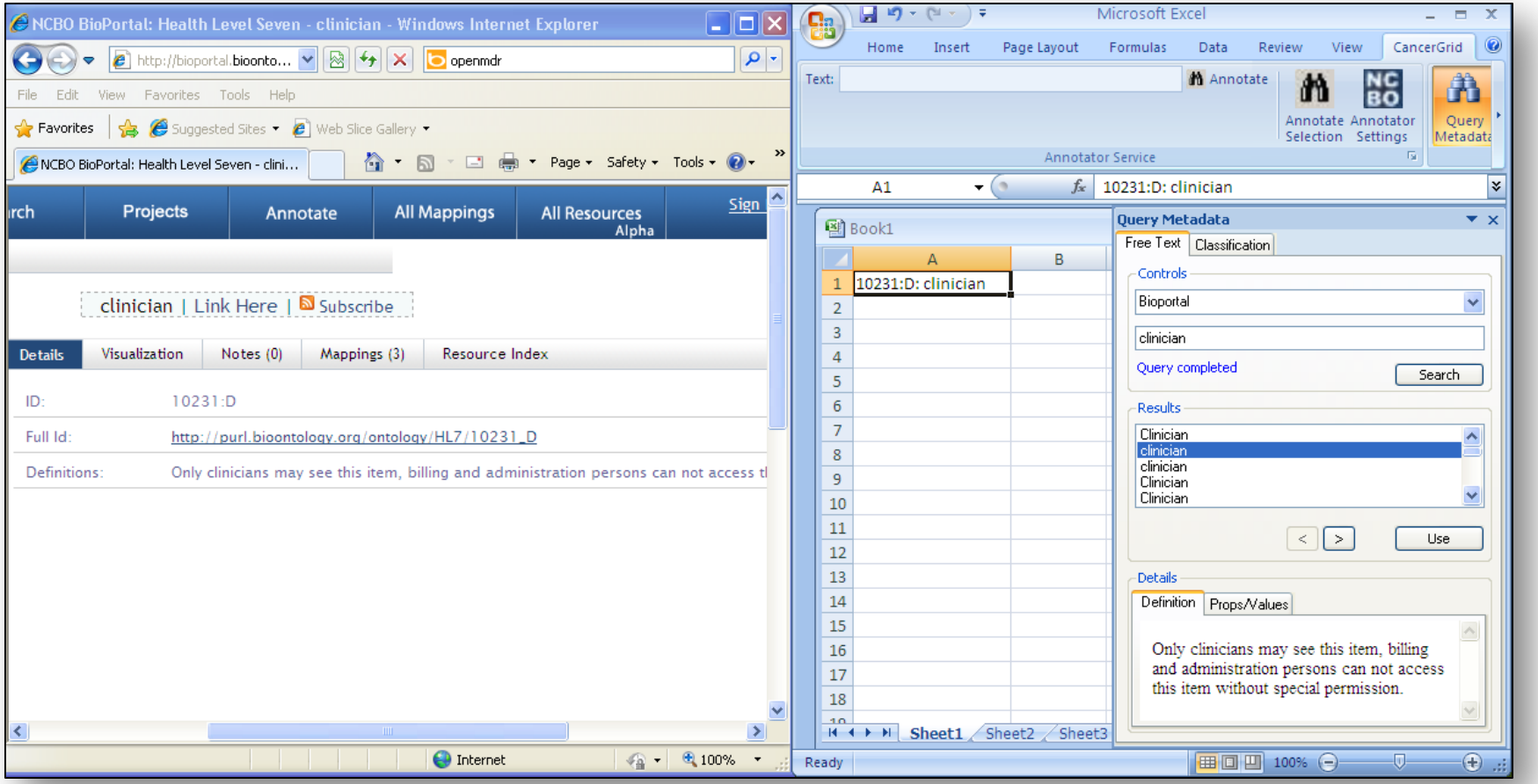
DOG4DAG Ontology Generation Tool for OBO-Edit and Protégé



Word Add-in

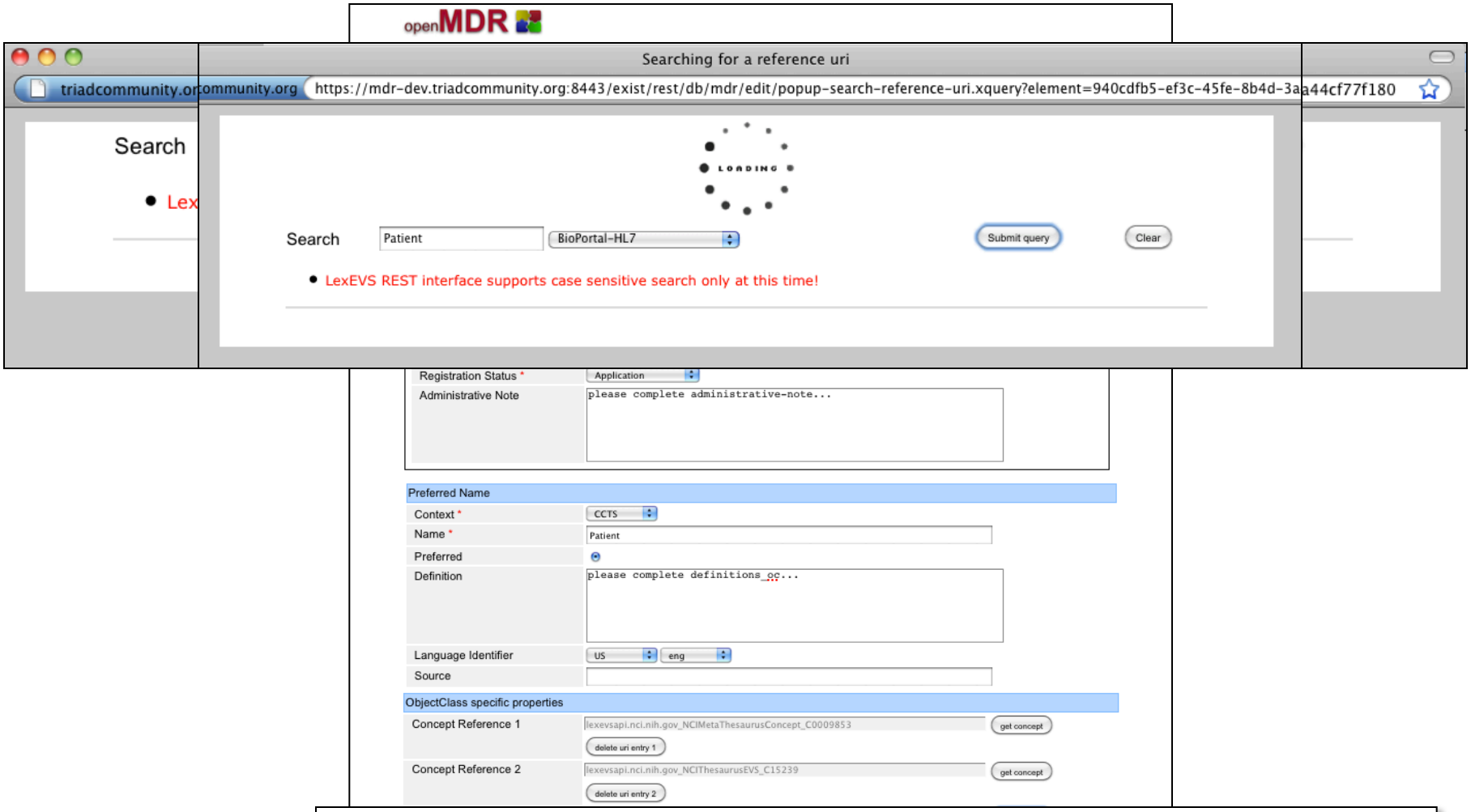

test_manuscript - Microsoft Word
Home
Insert
Page Layout
References
Mailings
Review
View
Developer
BioLit
Turn Parsing Off
Undo Highlighting
Configure Bio-Lit
BioLit Ontology Recognition Application
InfoPane
BioLit Ontology Recognition Application
Background
Tuberculosis (TB) remains a major cause of mortality in developing countries, and in these countries diabetes prevalence is increasing rapidly. Diabetes increases the risk of TB. Our aim was to assess the potential impact of diabetes as a risk factor for incident pulmonary tuberculosis, using India as an example.
Methods
We constructed an epidemiological model using data on tuberculosis incidence, diabetes
BioLit: prevalence
Add mark-up for ID:0000019 - quality of host population(Infectious disease ontology)
Remove this Smart Tag
Stop Recognizing "prevalence"
Smart Tag Options...
Results
In India in 2000 there were an estimated 20.7 million adults with diabetes, and 900,000 incident adult cases of pulmonary tuberculosis. Our calculations suggest that diabetes accounts for 14.8% (uncertainty range 7.1% to 23.8%) of pulmonary tuberculosis and 20.2% (8.3% to 41.9%) of smear-positive (i.e. infectious) tuberculosis.
We estimate that the increased diabetes prevalence in urban areas is associated with a 15.2% greater smear-positive tuberculosis incidence in urban than rural areas – over a fifth of the
Page: 1 of 2
Words: 244
Biolit Info Pane
BROWSER
INFO
incidence
INFECTIOUS DISEASE ONTOLOGY
quality
quality of infectious disease
incidence
INFECTIOUS DISEASE ONTOLOGY
quality
quality of infectious disease
incidence
Ontology Name:
100%

cgMDR



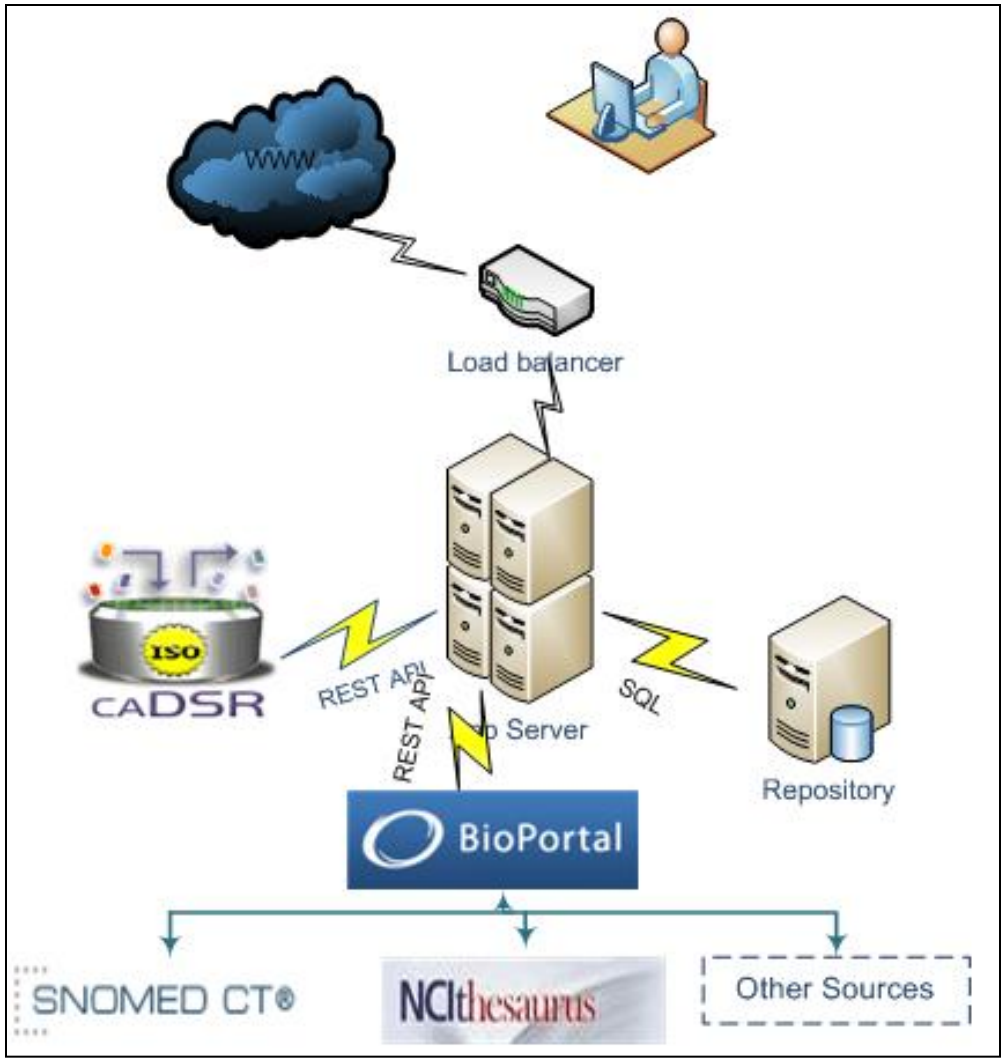
http://cancergrid.org/index.php

openMDR



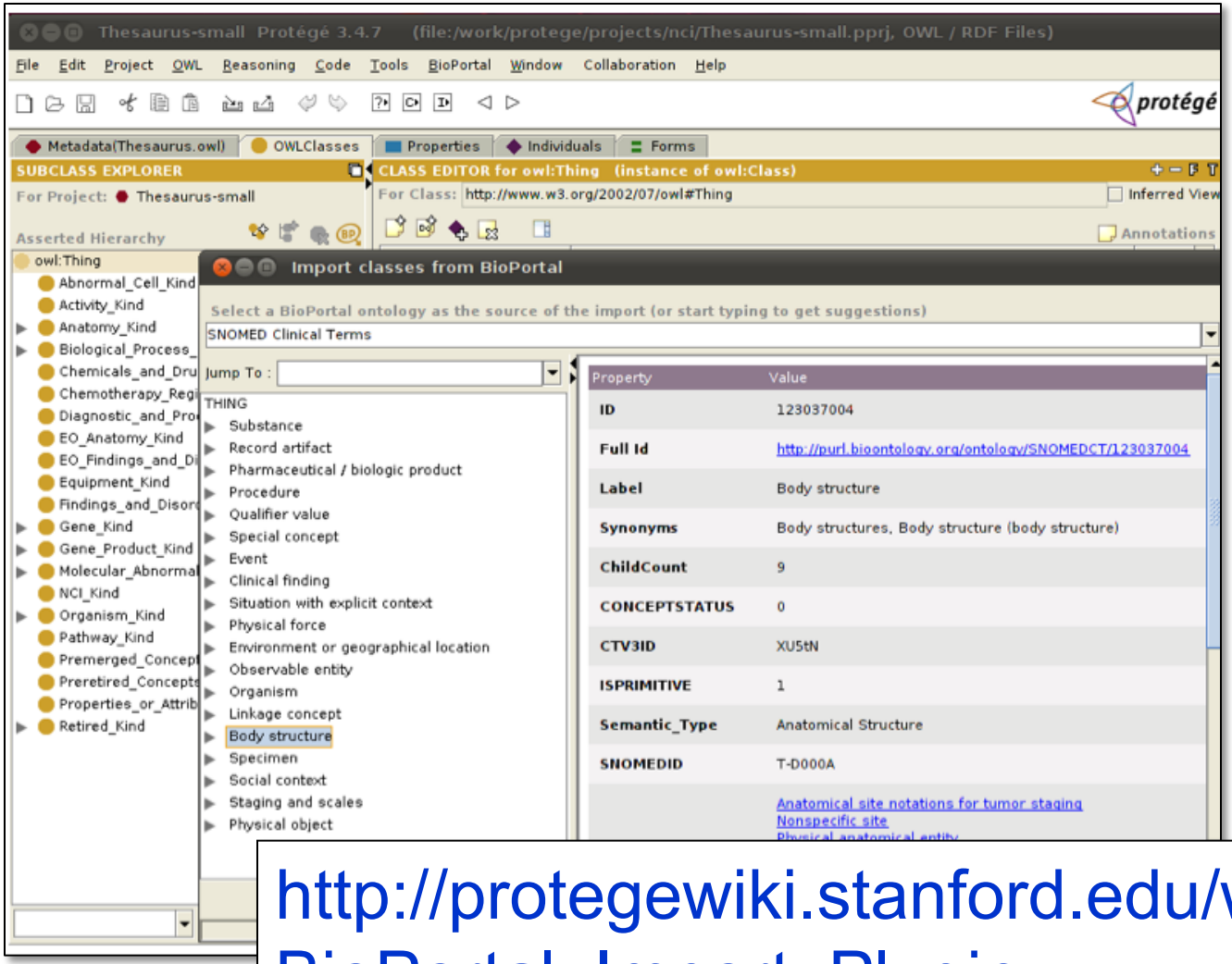
http://citih.osumc.edu/projects/project&r=1032

eleMap

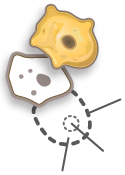


https://victr.vanderbilt.edu/eleMAP/

BioPortal Import Plugin

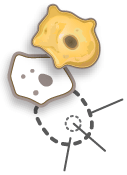


http://protegewiki.stanford.edu/wiki/BioPortal_Import_Plugin



NOTES WEB SERVICES

Propose new terms and comment on ontologies



Notes Web Service

- Add terms proposals and comments on ontology terms
- Documentation: http://www.bioontology.org/wiki/index.php/Ontology_Notes

Biomedical Resource Ontology

Notes

Notes

Subscribe to notes emails

Show 50 entries Filtering Options » Search:

SUBJECT	AUTHOR	TYPE	TARGET	CREATED
New Term Proposal: Exercise Study Facility archived	Mette	New Term Proposal	Physiology Facility (Class)	06/15/2010

New Term Proposal: Exercise Study Facility archived

New Term Proposal submitted by Mette over 1 year ago on Physiology Facility in Biomedical Resource Ontology

PREFERRED NAME	Exercise Study Facility	PROVISIONAL ID		PARENT	http://bioontology.org/ontologies/BiomedicalResourceOntology.owl#Physiology_Facility
REASON FOR CHANGE	Physiology facility child	STATUS		CONTACT INFO	
SYNONYMS					
DEFINITION	A facility or core devoted to exercise studies				

Responses

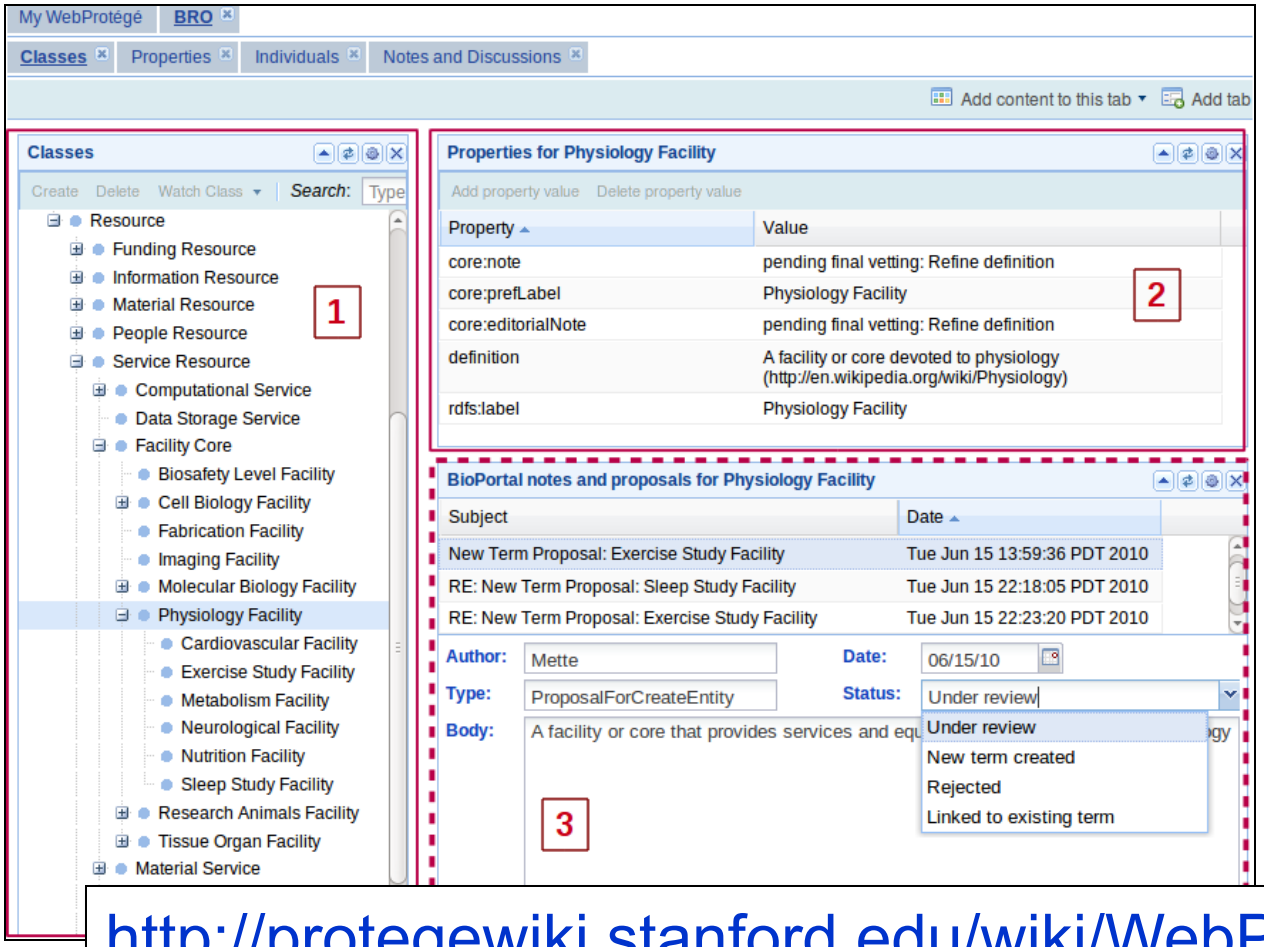
hide all | show all

RE: New Term Proposal: Exercise Study Facility by whetzel over 1 year ago

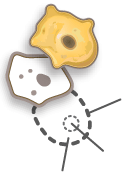
Can you expand on what an exercise study is?

reply

WebProtégé

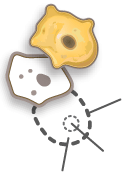


http://protegewiki.stanford.edu/wiki/WebProtege



VIEWS

Custom subset of large ontologies



Views and Value Sets

- Users can submit their own derivatives of BioPortal ontologies
  - these which become first-class objects in BioPortal and can be used as all other Web services

Views in BioPortal

Views

Create new view

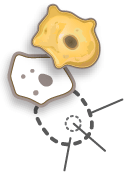
Expand All | Collapse All

- CORE Subset of SNOMED CT
- Neoplasm_breast_cancer
- Neoplasm_hamartoma
- SNOMED Anatomy
- SNOMED Clinical Findings

  - Description: The Clinical Finding subtree of SNOMED CT
  - Definition: Class subtree of ClinicalFinding
  - Ontology ID: 2018
  - Definition Language: Manual

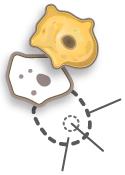
VERSION	BASE VERSION	CREATED	CREATED BY	ONTOLOGY FILE	DIFF FILE	VISIBILITY
1.2	2009_07_31	07/16/2010	Tania Tudorache, tudorache@stanford.edu	Download View		Public
1.1	2009_07_31	03/23/2010	Tania Tudorache, tudorache@stanford.edu	Download View		Public
1.0	2009_01_31	09/09/2009	Tania Tudorache, tudorache@stanford.edu	Download View		Public

- SNOMED Ethnic Group
- SNOMED Morphologically Abnormal Structure
- SNOMED Organism
- SNOMED Terminos Clinicos
- SNOMED Test Findings



Views and Value Sets

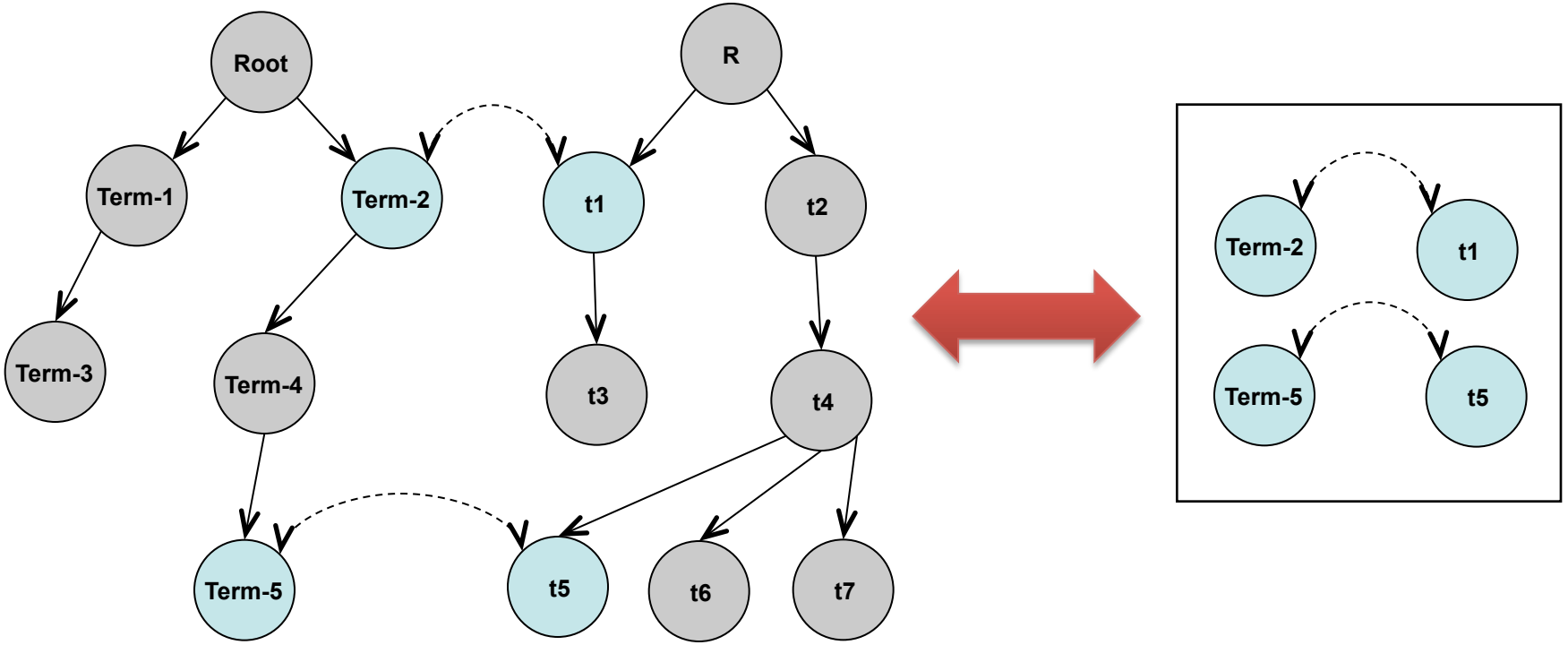
- View Extraction Web service
  - Given a root node term, extracts all child terms http://bit.ly/uXeh2s
- Access directly from Protégé via the BioPortal Import plugin http://protegewiki.stanford.edu/wiki/BioPortal_Import_Plugin



MAPPING WEB SERVICES

Using NCBO technology to integrate terminologies and ontologies

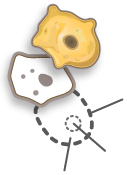
Mappings



Ontology A

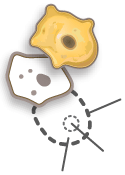
Ontology B

Upload or Download mapping subsets



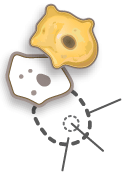
Mappings

- Mappings Web service
  - Documentation: http://www.bioontology.org/wiki/index.php/BioPortal_Mappings_Service
- Functions
  - Get
  - Create/Upload
- Example Perl client to Get mappings: http://bit.ly/tDKPQd



WIDGETS

Using NCBO technology on your web pages

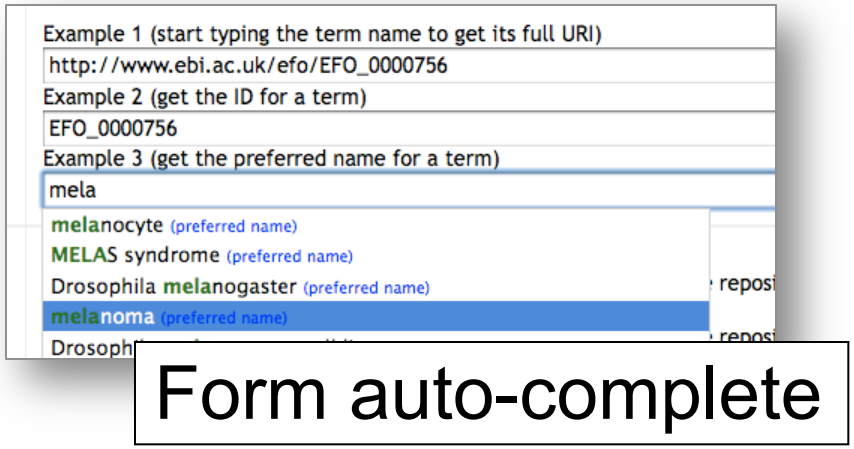


Widgets

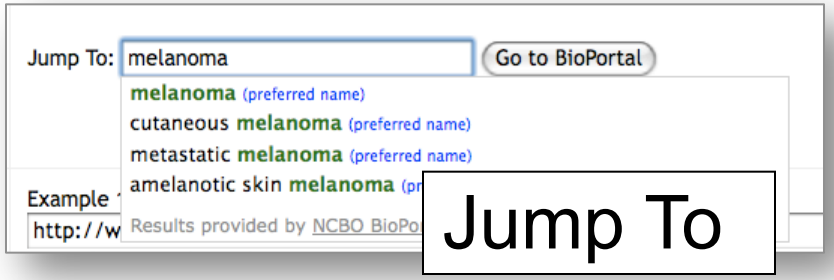
- Form auto-complete – auto-complete function and can return term URI, term ID or term name
- Jump To – auto-complete function to select term and Jump To BioPortal to view term details
- Visualize widget – view the ontology structure and relations
- Tree widget – view the ontology tree

Code for widgets is available on BioPortal

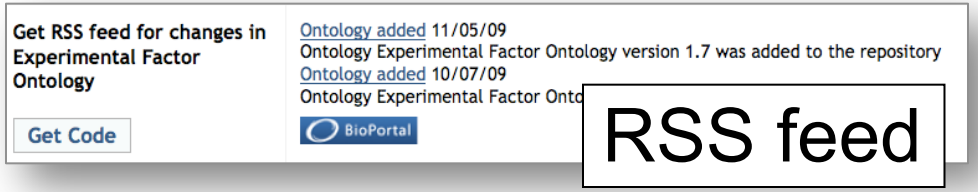
Widgets



Form auto-complete

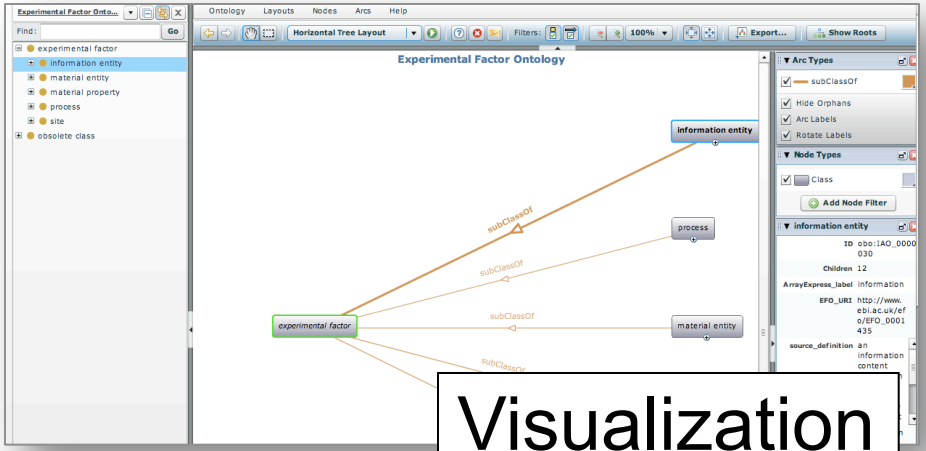


Jump To

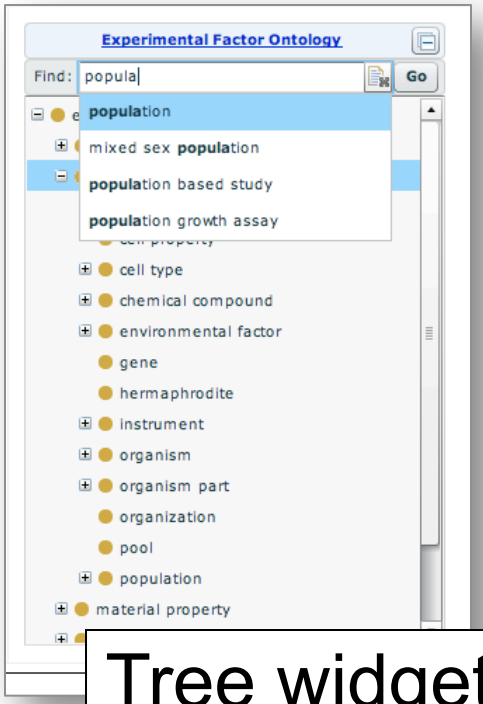


RSS feed

Widgets

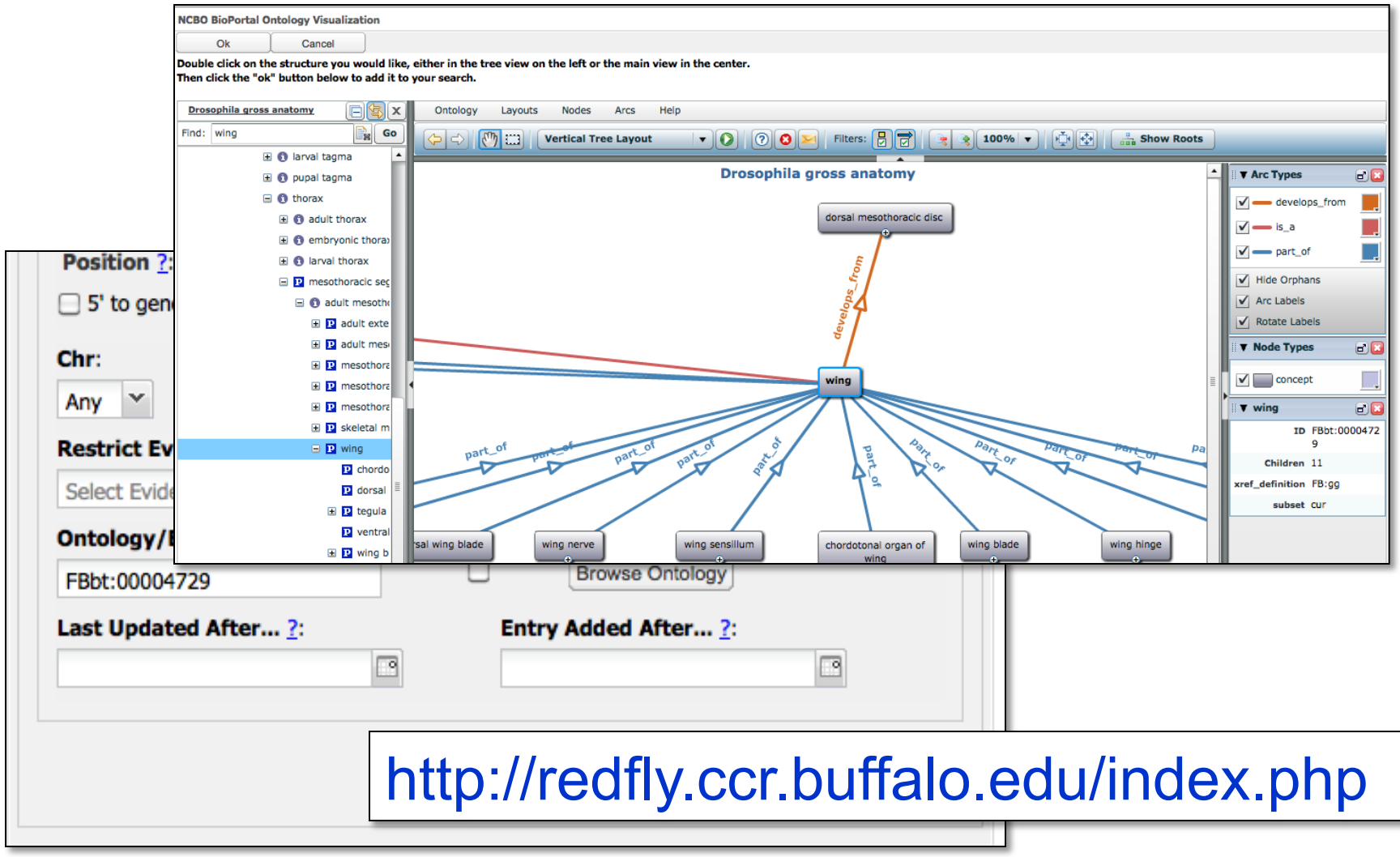


Visualization



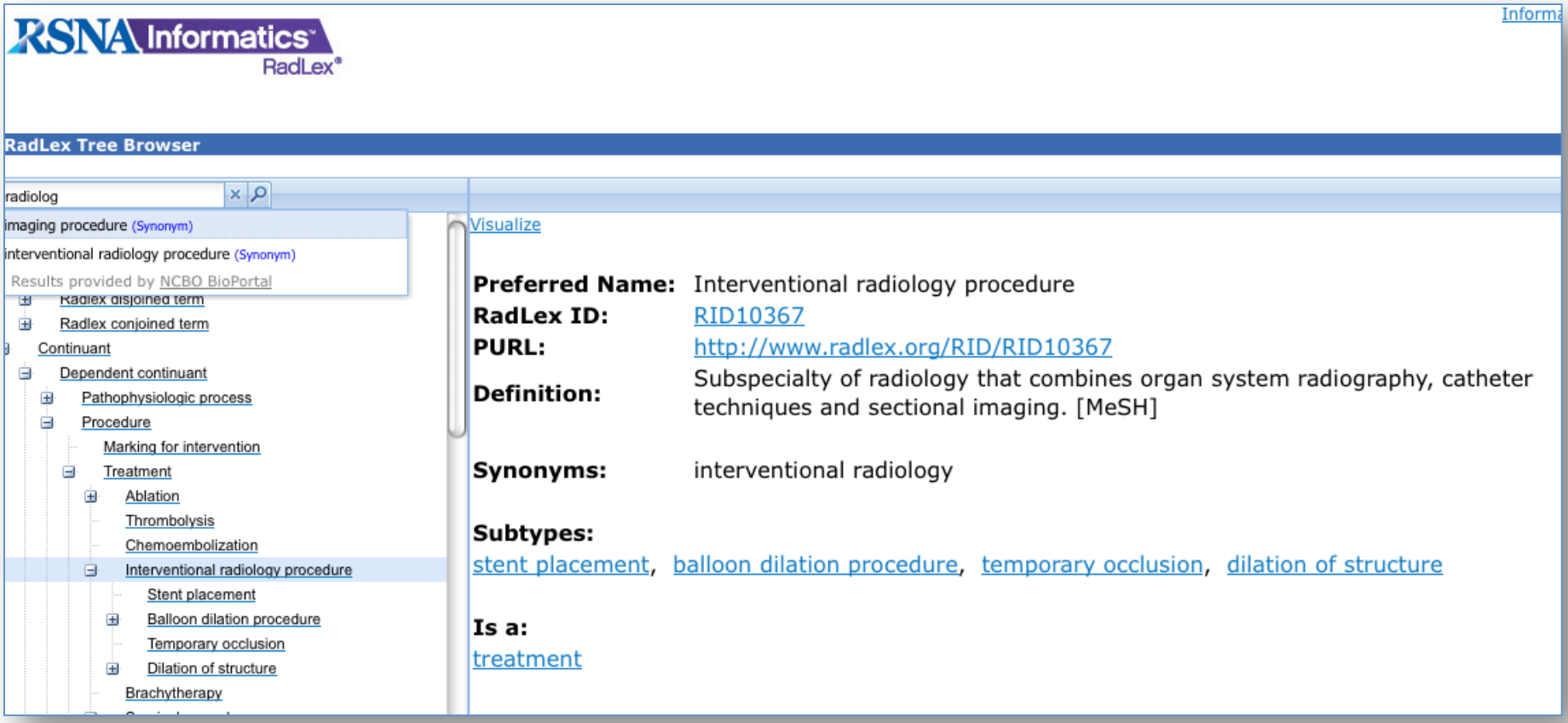
Tree widget

RedFly



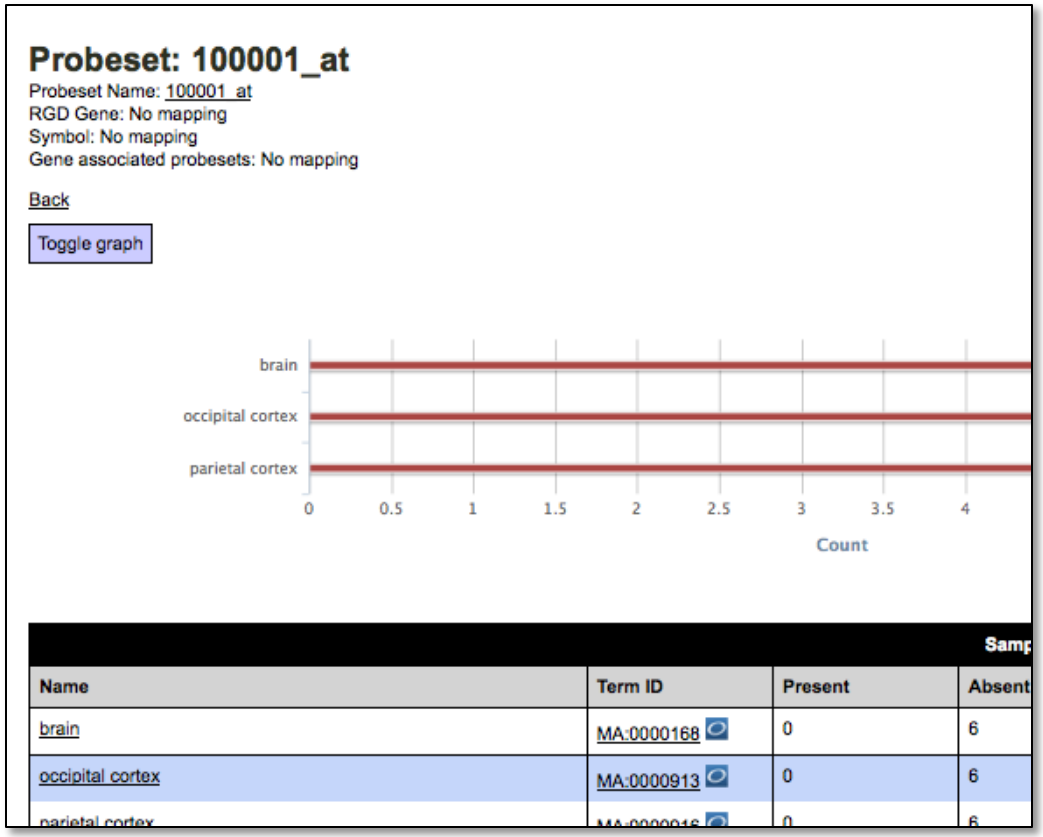
http://redfly.ccr.buffalo.edu/index.php

Radiological Society of North America



http://radlex.org/

GMiner

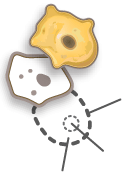


http://gminer.mcw.edu/

Knowledge Egg



http://www.kunnskapsegget.no/

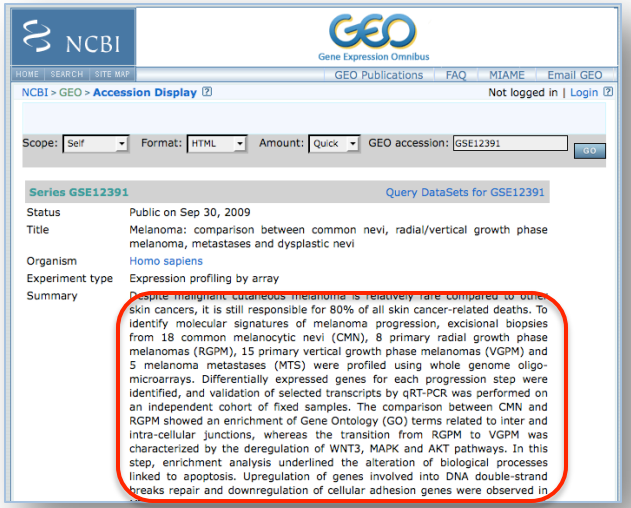


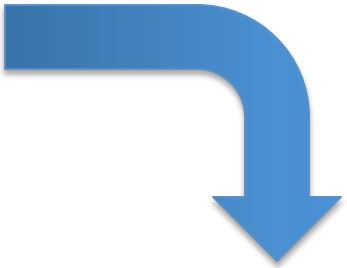
ANNOTATOR WEB SERVICE

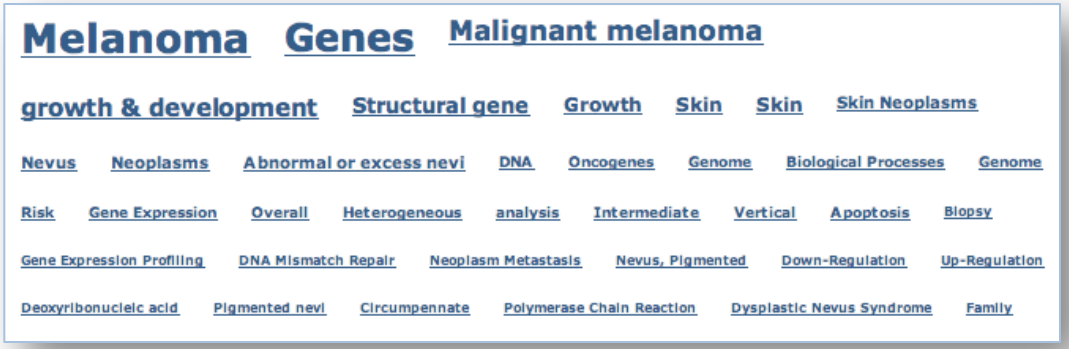
Using Ontologies to Annotate *Your* Data

Annotator: The Basic Idea

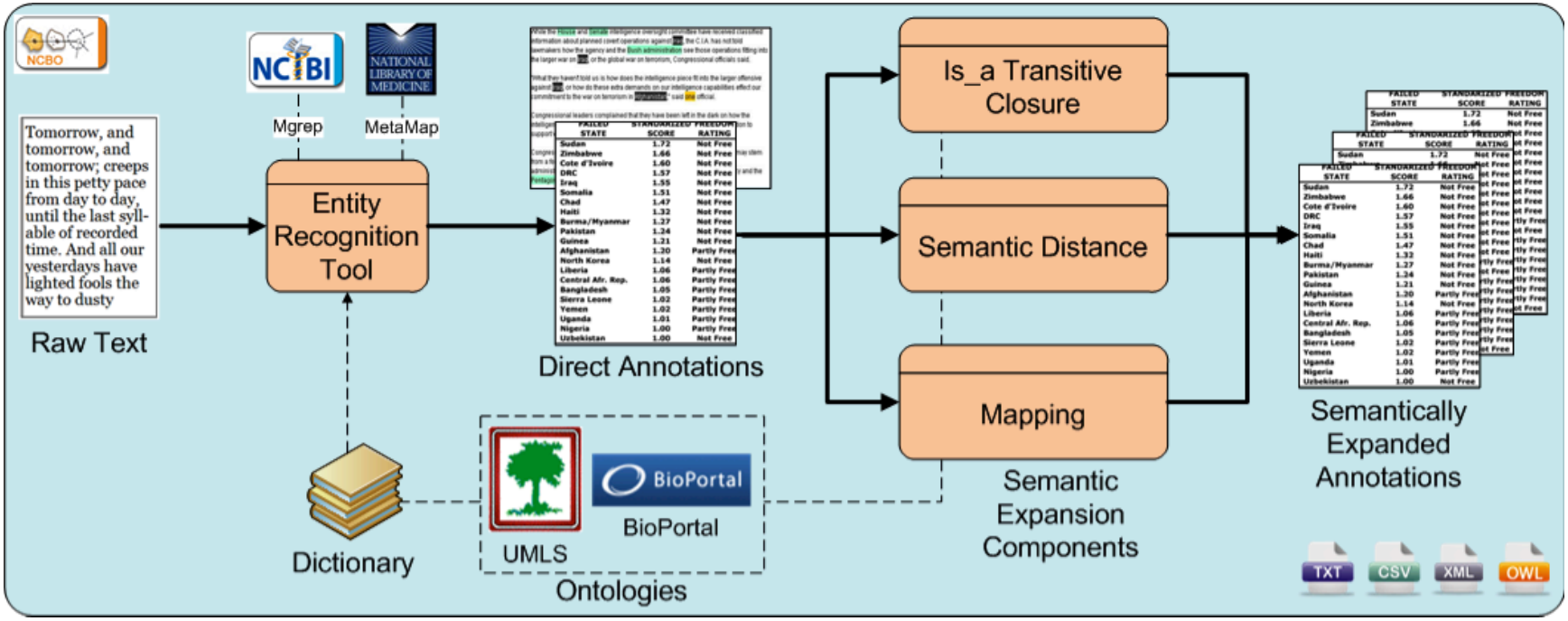
- Tag textual metadata with ontology terms

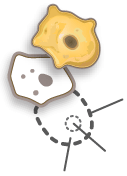






Annotator Workflow

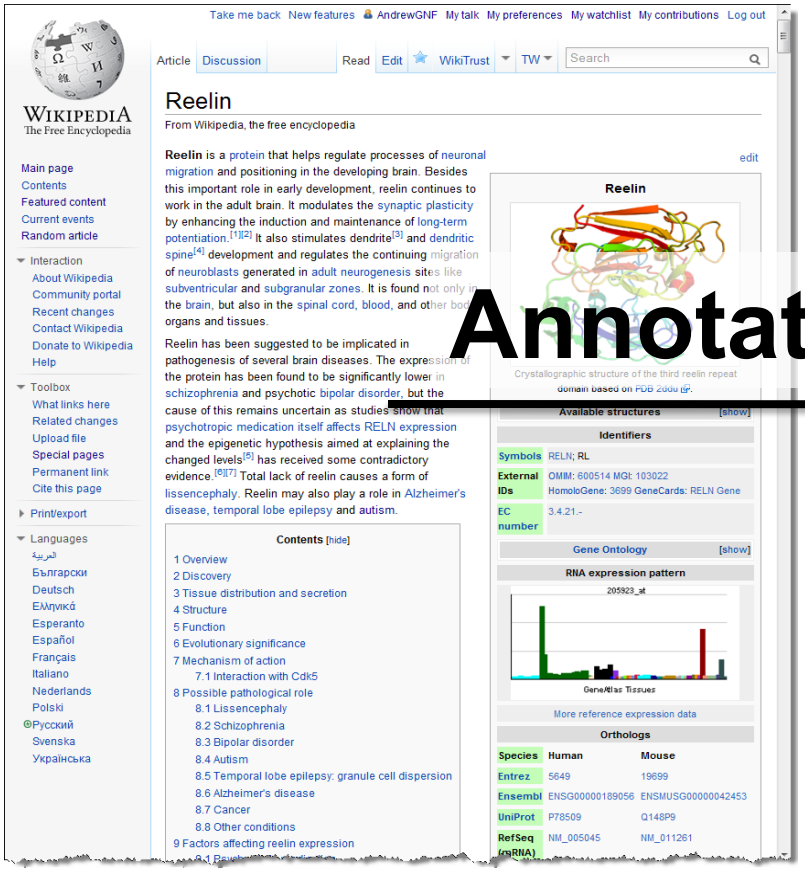




Annotator Web Service

- Tag free text with ontology terms
- Documentation: http://www.bioontology.org/wiki/index.php/Annotator_Web_service
- Example clients:
  - Perl: http://bit.ly/vKYpCP
  - Java: http://bit.ly/rRFoKd

Gene Wiki

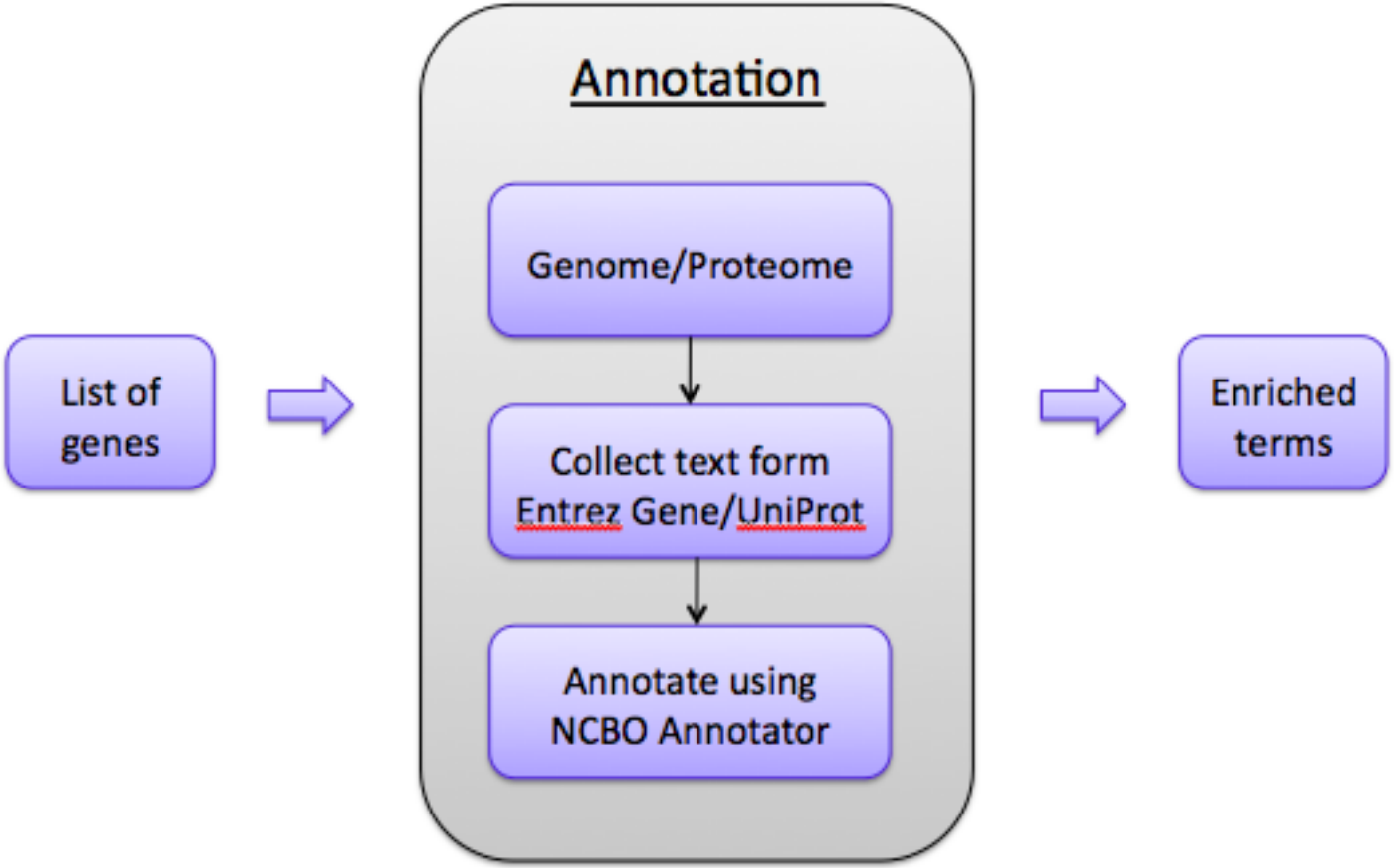


Annotator

Gene	Property	Value
RELN	GO: Biological Process	Brain Development
RELN	DO: Related Disease	Alzheimer's

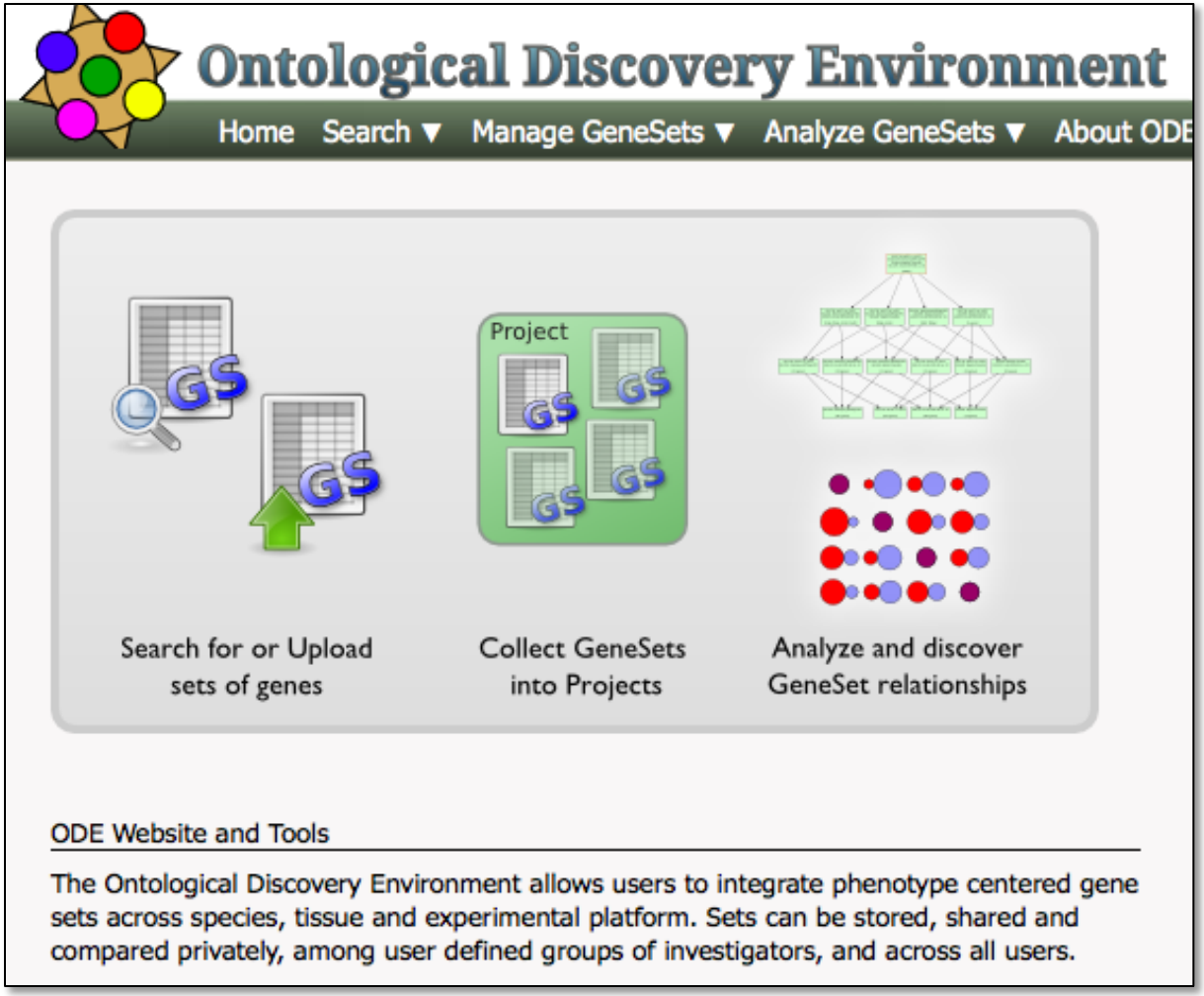
http://en.wikipedia.org/wiki/Portal:Gene_Wiki

Statistical Tracking of Ontological Phrases (STOP)



http://www.mooneygroup.org/content/webtools

Ontological Discovery Environment



http://ontologicaldiscovery.org/

STRIDE

De-ided version of Note_id: 1058133

PATIENT NAME: X

ADMISSION DIAGNOSIS: Cervical spondylosis at C5-C6 with disk hemiation on the left side and left-sided C6 radiculopathy.

DISCHARGE DIAGNOSIS: Cervical spondylosis at C5-C6 with disk hemiation on the left side and left-sided C6 radiculopathy.

PROCEDURES PERFORMED: C5-C6 anterior cervical diskectomy and fusion with instrumentation, performed by Y

REASON FOR HOSPITALIZATION: X is a 48-year-old man with left-sided neck pain and radiation down to the left side of his shoulder and upper arm with numbness of the thumb, index and middle fingers at the tips, more on the left, and ongoing for the past three months. The patient had an MRI scan of the cervical spine done in early MONTH, which revealed cervical spondylosis at C5-C6 with disk herniation.

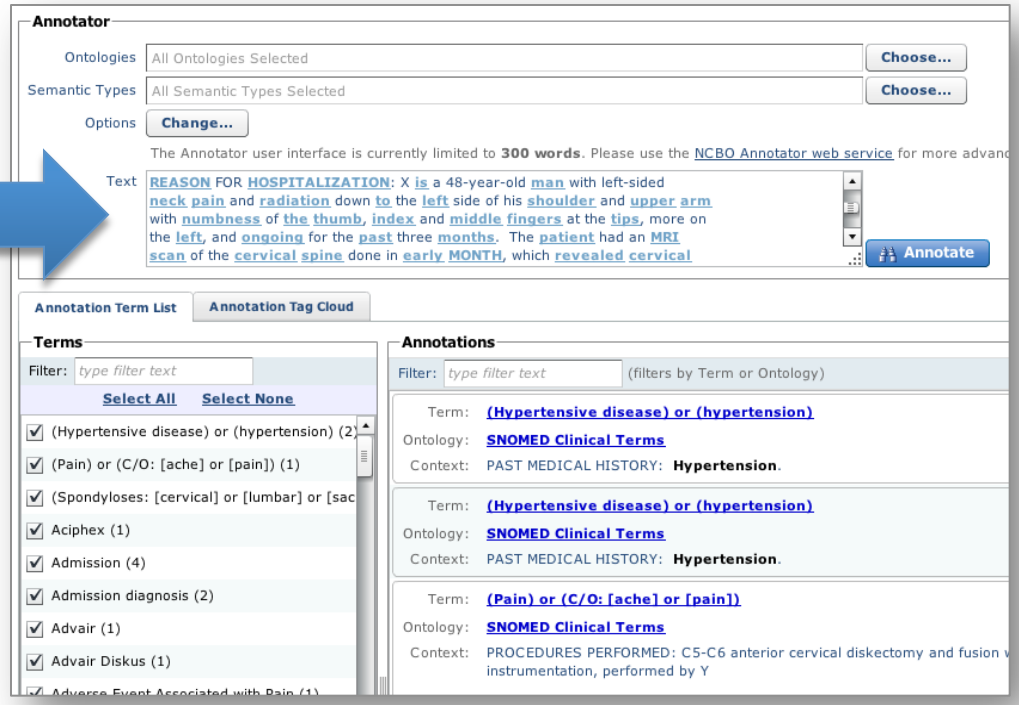
PAST MEDICAL HISTORY: Hypertension.

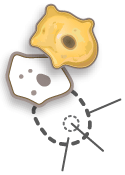
PAST SURGICAL HISTORY: None.

MEDICATIONS ON ADMISSION:
1. Advair Diskus one puff b.i.d.
2. Aciphex 20 mg one tablet q.a.m.
3. Flovent MDI two puffs p.o. b.i.d.
4. Albuterol MDI two puffs q.4-6h. p.r.n.
5. Flonase two sprays per nostril q.d.
6. Nexium 40 mg one capsule q.d.

PHYSICAL EXAMINATION ON ADMISSION: GENERAL: The patient had clear consciousness and no acute distress. NEUROLOGIC: The patient's range of motion was 70% of normal, but there was no Lhermitte sign and no Spurling sign. There were no focal weaknesses present for both upper and lower extremities. Brachioradialis was 1/4 on both sides. Triceps was 1/4 to 2/4 on both sides. Knee jerk and ankle jerk were 2/4 on both sides. Posterior tibialis was 2/4 on both sides. LUNGS: Clear to auscultation bilaterally. CARDIOVASCULAR: Regular rate and rhythm. ABDOMEN: Benign. EXTREMITIES: No cyanosis or edema.

HOSPITAL COURSE: The patient was admitted to the hospital on 2003, and taken to the operating room for a C5-C6 ACDF with instrumentation. The patient tolerated the procedure well. There were no complications. The patient recovered well in the PACU without any events, and the patient was transferred to the floor. While on the floor there were no acute events overnight. The patient was tolerating oral intake, and the patient's pain was under control with oral pain medications. On postoperative day #1 the patient was tolerating an oral diet, ambulating and voiding on his own, and the patient was discharged home.

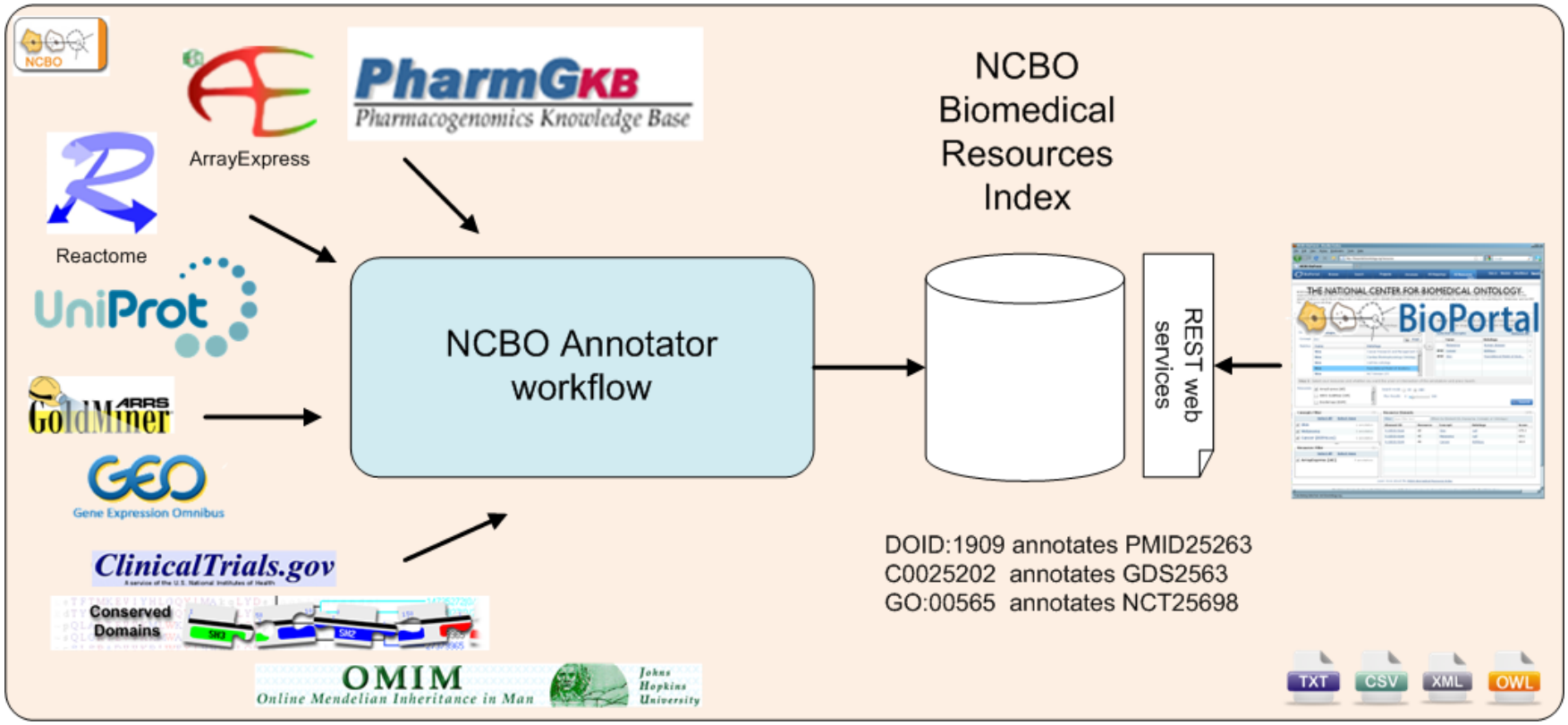




DATA SERVICE

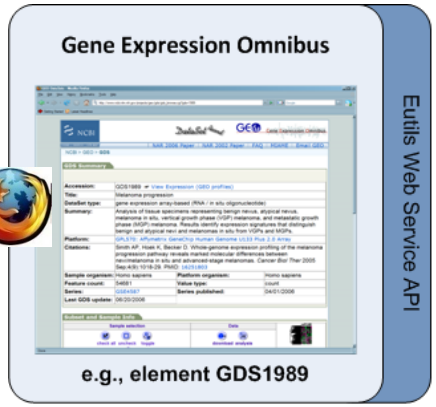
Using Ontologies to Access Public Data

Resource Index: The Basic Idea



- The index can be used for
  - Search
  - Data mining

Resources index: Example



Accessing resource elements

<title>
Melanoma progression
</title>

<description>
Analysis of tissue specimens representing benign nevus, atypical nevus, melanoma in situ, vertical growth phase (VGP) melanoma, and metastatic growth phase (MGP) melanoma. Results identify expression signatures that distinguish benign and atypical nevi and melanomas in situ from VGPs and MGPs.
</description>

Concept recognition

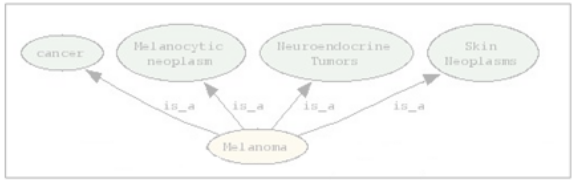
<title>
Melanoma progression
</title>

<description>
Analysis of tissue specimens representing benign nevus, atypical nevus, melanoma in situ, vertical growth phase (VGP) melanoma, and metastatic growth phase (MGP) melanoma. Results identify expression signatures that distinguish benign and atypical nevi and melanomas in situ from VGPs and MGPs.
</description>

23 direct annotations
(4 title, 19 description)

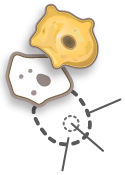
Example:
Melanoma, concept (DOID:1909) in ontology *Human disease*.

Semantic expansion



172 closure annotations

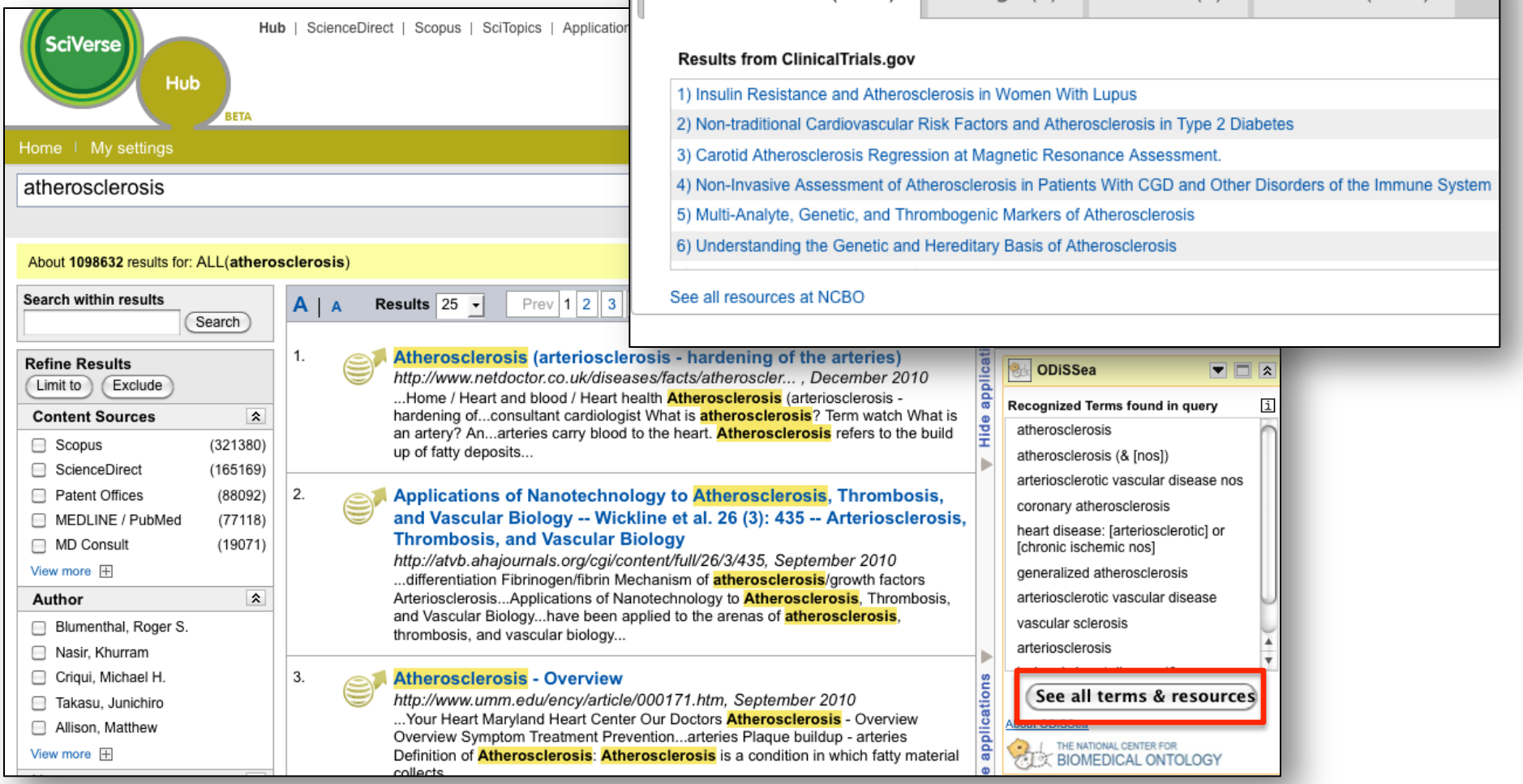
Examples:
Cancer, concept (DOID:162) in ontology *Human disease*
Skin Neoplasms, concept (DOID:3165) in ontology *Human disease*

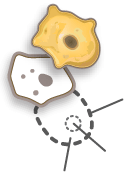


Data Access

- Resource Index Web service
  - Documentation: http://www.bioontology.org/wiki/index.php/Resource_Index

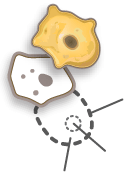
ODiSSea


Clinical Trials (1146)
Drugs (2)
Genes (3)
Grants (1147)
Results from ClinicalTrials.gov
1) Insulin Resistance and Atherosclerosis in Women With Lupus
2) Non-traditional Cardiovascular Risk Factors and Atherosclerosis in Type 2 Diabetes
3) Carotid Atherosclerosis Regression at Magnetic Resonance Assessment.
4) Non-Invasive Assessment of Atherosclerosis in Patients With CGD and Other Disorders of the Immune System
5) Multi-Analyte, Genetic, and Thrombogenic Markers of Atherosclerosis
6) Understanding the Genetic and Hereditary Basis of Atherosclerosis
See all resources at NCBO
SciVerse
Hub
BETA
Hub | ScienceDirect | Scopus | SciTopics | Application
Home | My settings
atherosclerosis
About 1098632 results for: ALL(atherosclerosis)
Search within results
Search
Refine Results
Limit to
Exclude
Content Sources
Scopus (321380)
ScienceDirect (165169)
Patent Offices (88092)
MEDLINE / PubMed (77118)
MD Consult (19071)
View more
Author
Blumenthal, Roger S.
Nasir, Khurram
Criqui, Michael H.
Takasu, Junichiro
Allison, Matthew
View more
Results 25
Prev 1 2 3
1. Atherosclerosis (arteriosclerosis - hardening of the arteries)
http://www.netdoctor.co.uk/diseases/facts/atheroscler... , December 2010
...Home / Heart and blood / Heart health Atherosclerosis (arteriosclerosis - hardening of...consultant cardiologist What is atherosclerosis? Term watch What is an artery? An...arteries carry blood to the heart. Atherosclerosis refers to the build up of fatty deposits...
2. Applications of Nanotechnology to Atherosclerosis, Thrombosis, and Vascular Biology -- Wickline et al. 26 (3): 435 -- Arteriosclerosis, Thrombosis, and Vascular Biology
http://atvb.ahajournals.org/cgi/content/full/26/3/435, September 2010
...differentiation Fibrinogen/fibrin Mechanism of atherosclerosis/growth factors Arteriosclerosis...Applications of Nanotechnology to Atherosclerosis, Thrombosis, and Vascular Biology...have been applied to the arenas of atherosclerosis, thrombosis, and vascular biology...
3. Atherosclerosis - Overview
http://www.umm.edu/ency/article/000171.htm, September 2010
...Your Heart Maryland Heart Center Our Doctors Atherosclerosis - Overview Overview Symptom Treatment Prevention...arteries Plaque buildup - arteries Definition of Atherosclerosis: Atherosclerosis is a condition in which fatty material collects
Hide applicati
e applications
ODiSSea
Recognized Terms found in query
atherosclerosis
atherosclerosis (& [nos])
arteriosclerotic vascular disease nos
coronary atherosclerosis
heart disease: [arteriosclerotic] or [chronic ischemic nos]
generalized atherosclerosis
arteriosclerotic vascular disease
vascular sclerosis
arteriosclerosis
See all terms & resources
About ODiSSea
THE NATIONAL CENTER FOR BIOMEDICAL ONTOLOGY



BioPortal SPARQL Endpoint

- Documentation: http://www.bioontology.org/wiki/index.php/SPARQL_BioPortal
- Query interface: http://alphasparql.bioontology.org/
- Example queries: http://alphasparql.bioontology.org/examples
- Sample code: https://github.com/ncbo/sparql-code-examples



Thank you!

- Web service documentation:
  http://www.bioontology.org/wiki/index.php/NCBO_REST_services
- Keep in touch
  - NCBO Announce mailing list: http://bit.ly/osLzng
  - Software support: support@bioontology.org
  - Twitter: @bioontology
  - Facebook and LinkedIn
- Questions:
  support@bioontology.org